State of Crypto 2019

Blockchain Capital

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2019 Snapshot

1 | Macro Environment

2019 was marked by rising global debt levels and instability. The consequences are worrisome as quantitative easing becomes increasingly imminent. Simultaneously, youth protestors took to the streets fighting against political and economic inequality. Lastly, big tech forayed further into financial services. The unsteadiness of the macroeconomic environment, civil unrest and influx of new monopolistic players sets the stage for bitcoin.

2 | Crypto Assets

The crypto asset market recovered in 2019, up ~56% as of 11/30. Major token themes include the continued dominance of Tether among stablecoins and the rise, and subsequent, fall of IEOs. Conviction in bitcoin grew as the narrative shifted from store of value to potential "safe haven asset." Ethereum held its position as the leading smart contract platform, with "Decentralized Finance (DeFi)" emerging as the strongest use case.

3 | Adoption

Crypto assets are moving from fringe to mainstream across three segments: retail users, institutions and now, nation states. Digital assets continue to resonate with Gen Z / millennials. Institutional adoption pushed forward with both financial incumbents and big tech introducing products. Facebook's Libra was the most radical development. In lieu of Libra, several nations engaged with the concept of central bank digital currencies (CBDC).

4 | Regulation

U.S. regulation remains focused on two major areas: clarification of assets and transmission of value. Major cases included the Kik complaint, EOS settlement and an emergency restraining order against Telegram. FinCEN provided more helpful guidance regarding how the transmission of value via digital assets is regulated. On a global level, many nations leaned into the regulatory environment proposing new regulations and laws.

5 | Major Themes

DeFi was the biggest theme of 2019 with $600M+ in ETH locked in smart contracts. Lending emerged as the leading use case with Maker and Compound. The prime brokerage stack continues to develop with over a dozen players tackling custody, exchange and execution services. Gaming continues to be a focus for many base layer protocols. Lastly, developer tools, like node infrastructure and data analytics grew substantially.

6 | Looking Ahead

The industry is in its nascence, with a majority of the value captured today as part of the onboarding process. Over time, developer mindshare may shift from horizontal protocol competition to vertical construction up the stack as Layer 1 further ossifies. With impending regulation, widespread onramps and new use cases, the industry has potential to explode. Our team shares a few predictions for 2020.

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1

Macro Environment

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Macro environment sets the stage for crypto assets

Global macro uncertainty

- 2019 saw a continued flight to safety in a yield-starved climate with signs of imminent monetary easing
- Bitcoin is increasingly regarded as an uncorrelated safe-haven asset

Political unrest

- Leaderless movements took hold across the globe, questioning authoritarian motives and a wide range of social and fiscal policies

Big tech dominance

- Big Tech dominance grew, as many announced financial products (e.g., Apple Credit Card, Libra), placing a new category of data in their hands

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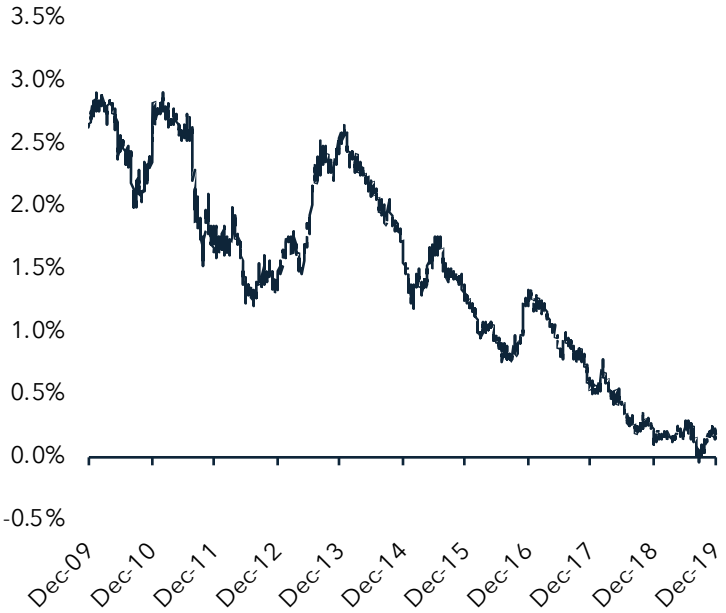
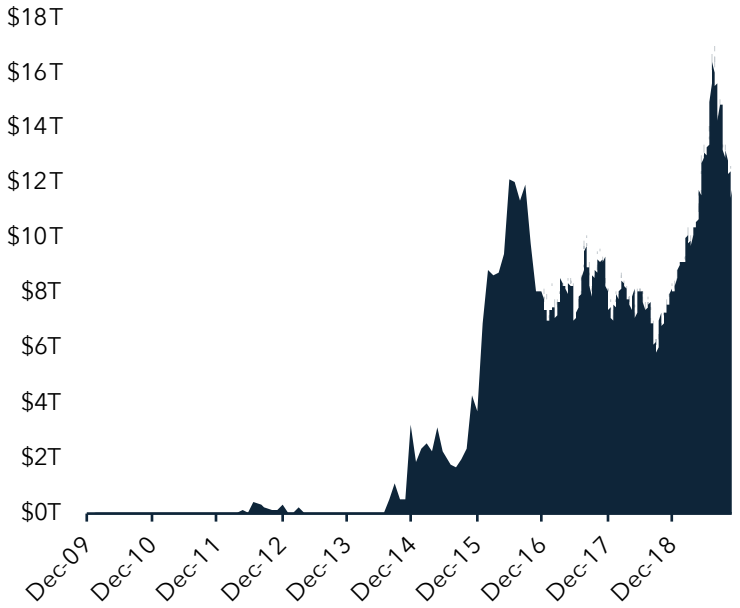
Flight to safety in a yield-starved climate

10-Year Treasury Constant Maturity Minus 2-Year Treasury Constant Maturity



Bloomberg Barclays Global Aggregate Negative-Yielding Debt Index



- Over the last decade, the Fed has pumped trillions of dollars into the economy, artificially inflating asset prices
- Global instability is on the rise, as evidenced by US-China trade tensions, slowing global growth and the increasing prevalence of populist narratives worldwide
- These factors (among others) have contributed to a flight to safety, and the combination of high asset values and low fixed income yields suggests allocators are in need of alternative stores of value

Source: FRED (Federal Reserve Bank of St. Louis), Bloomberg

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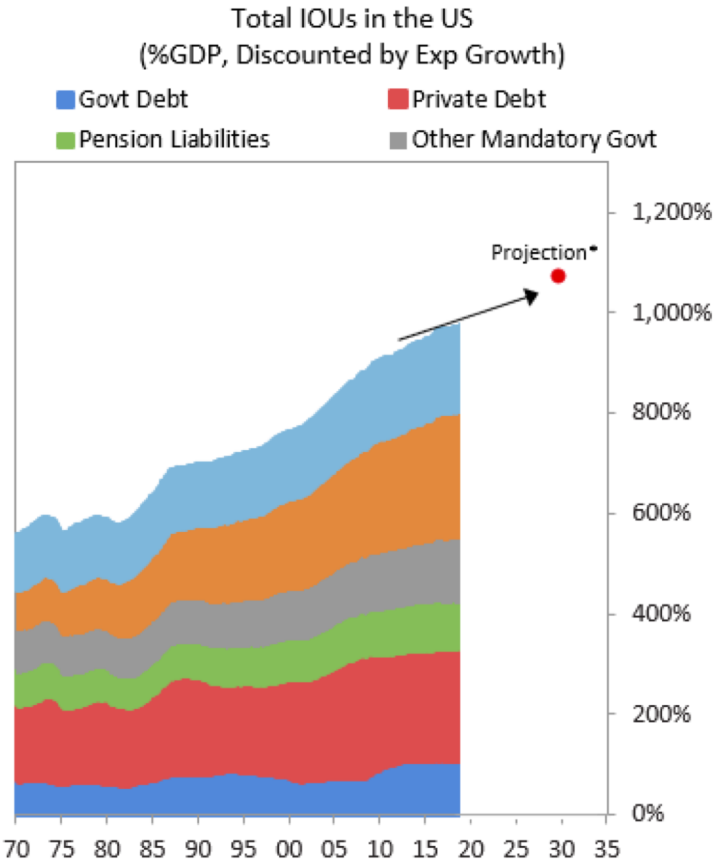
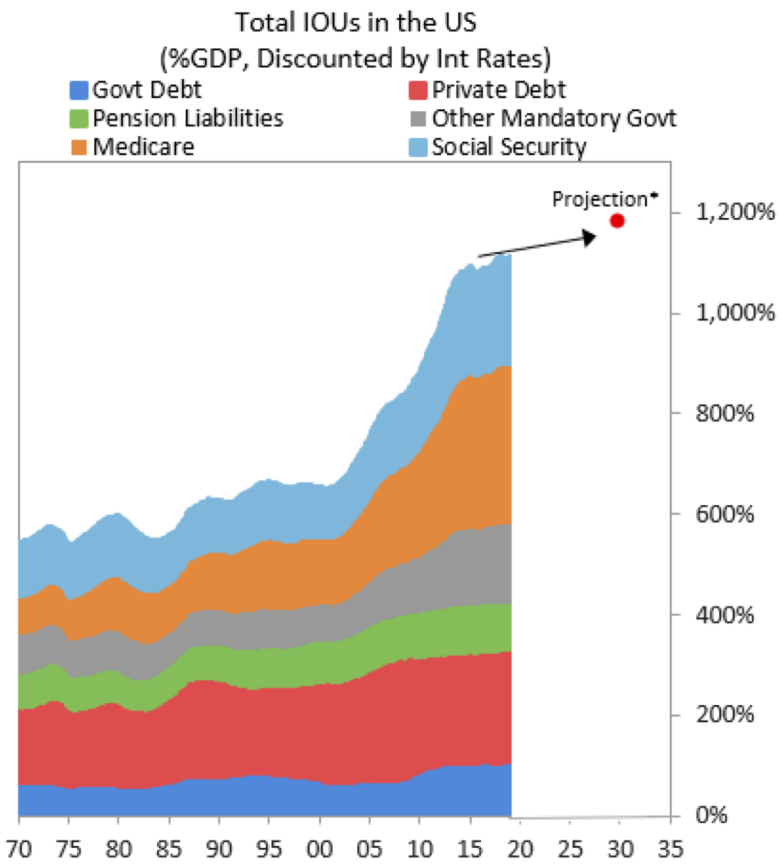
The future spells further monetary easing





Notwithstanding the growing debt burden, central banks are failing to hit inflation targets, suggesting monetary easing is imminent

Source: Bridgewater

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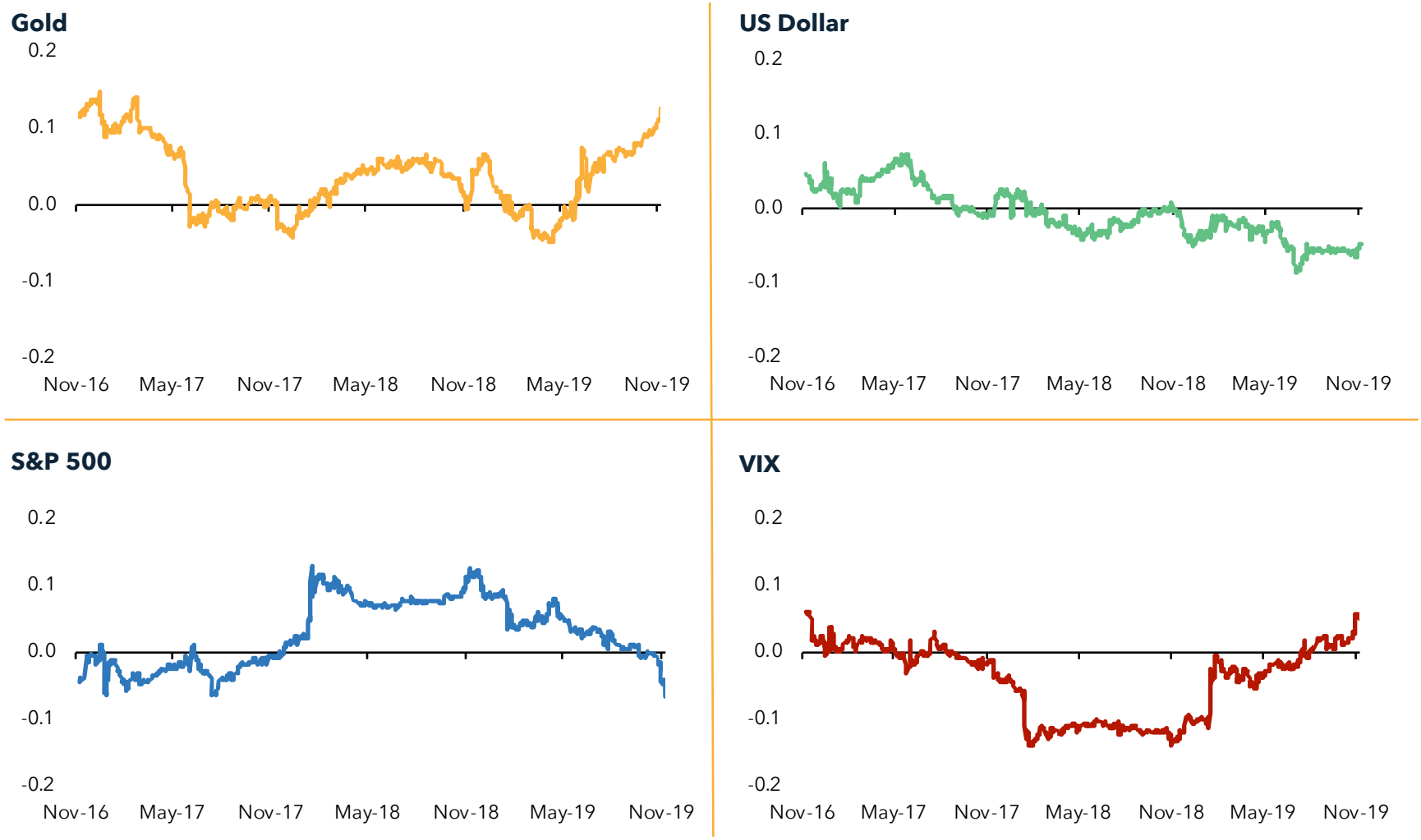
Enter bitcoin: an uncorrelated asset

One Year Rolling Correlations (Daily Log Returns)



Source: CoinMetrics, FRED (St. Louis Fed)

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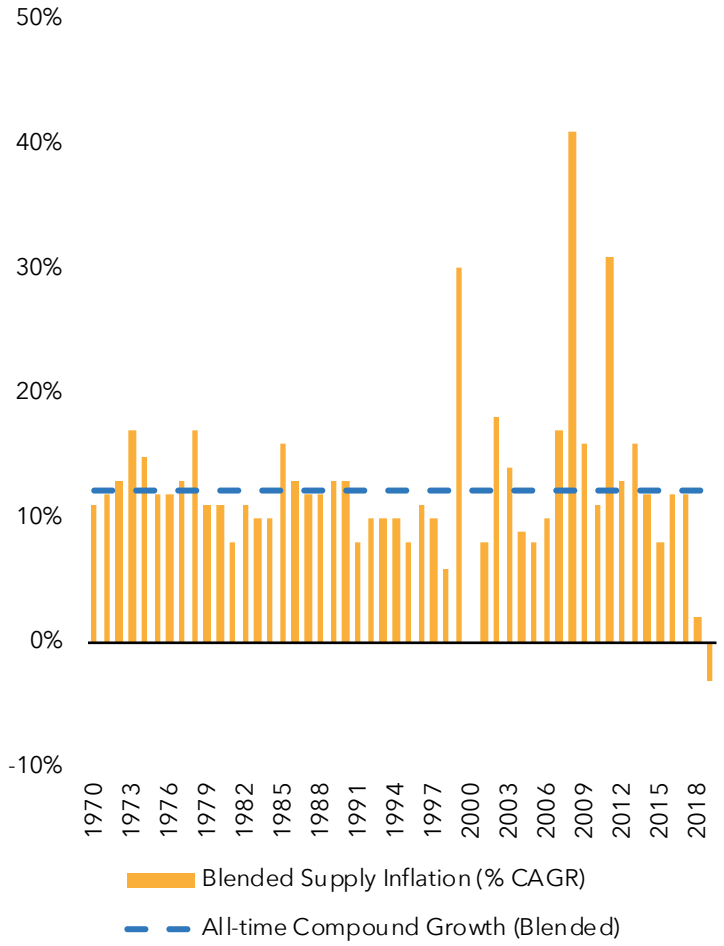
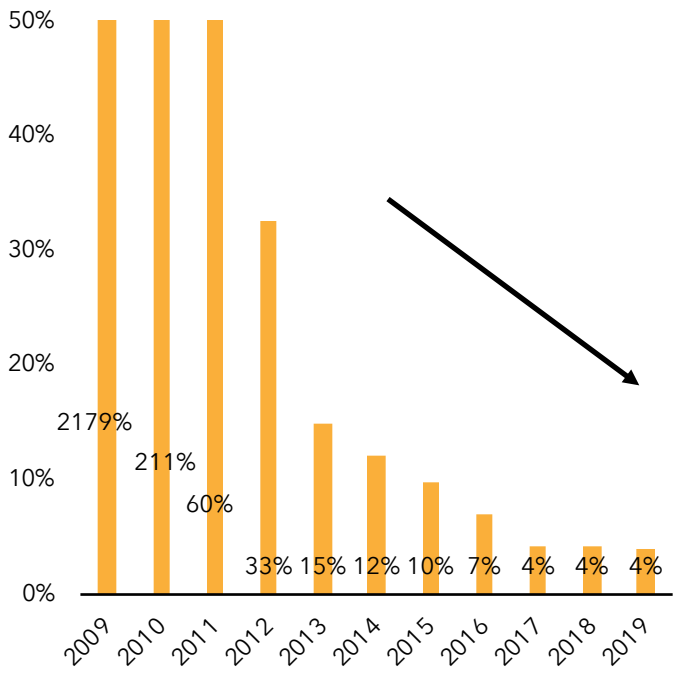
Base money: fiat vs bitcoin production rate

Fiat: Blended Supply Inflation (% CAGR)



Bitcoin Supply Inflation (% CAGR)



- Bitcoin's supply inflation halves every four years in perpetuity. The next halving event is May 2020.

Source: H/t to Crypto Voices (@crypto_voices) for the best data/analysis on base money production rates.

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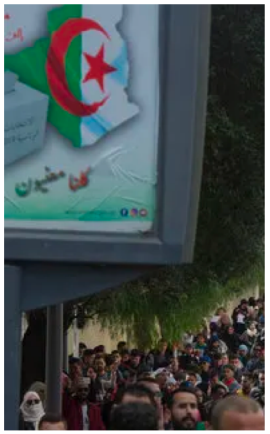
Civil protests reached new heights



Youth-led protests swept the world as people fought for democracy, financial equality and social justice

Source: Google Images

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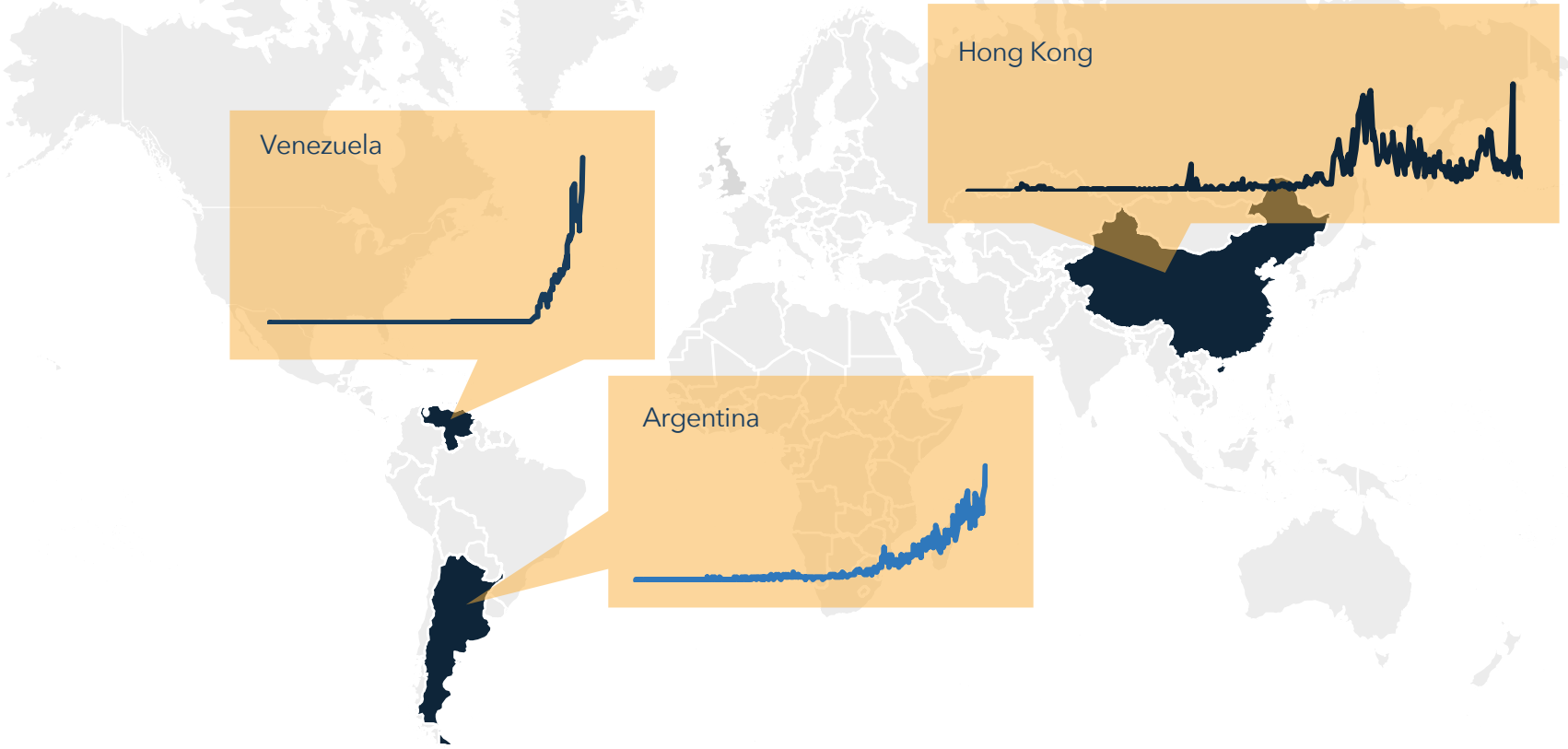
Bitcoin may be increasingly viewed as a political tool, decoupling money and state



In countries with civil unrest, 2019 LocalBitcoins volumes spiked

Source: LocalBitcoins, Coinshare Analysis

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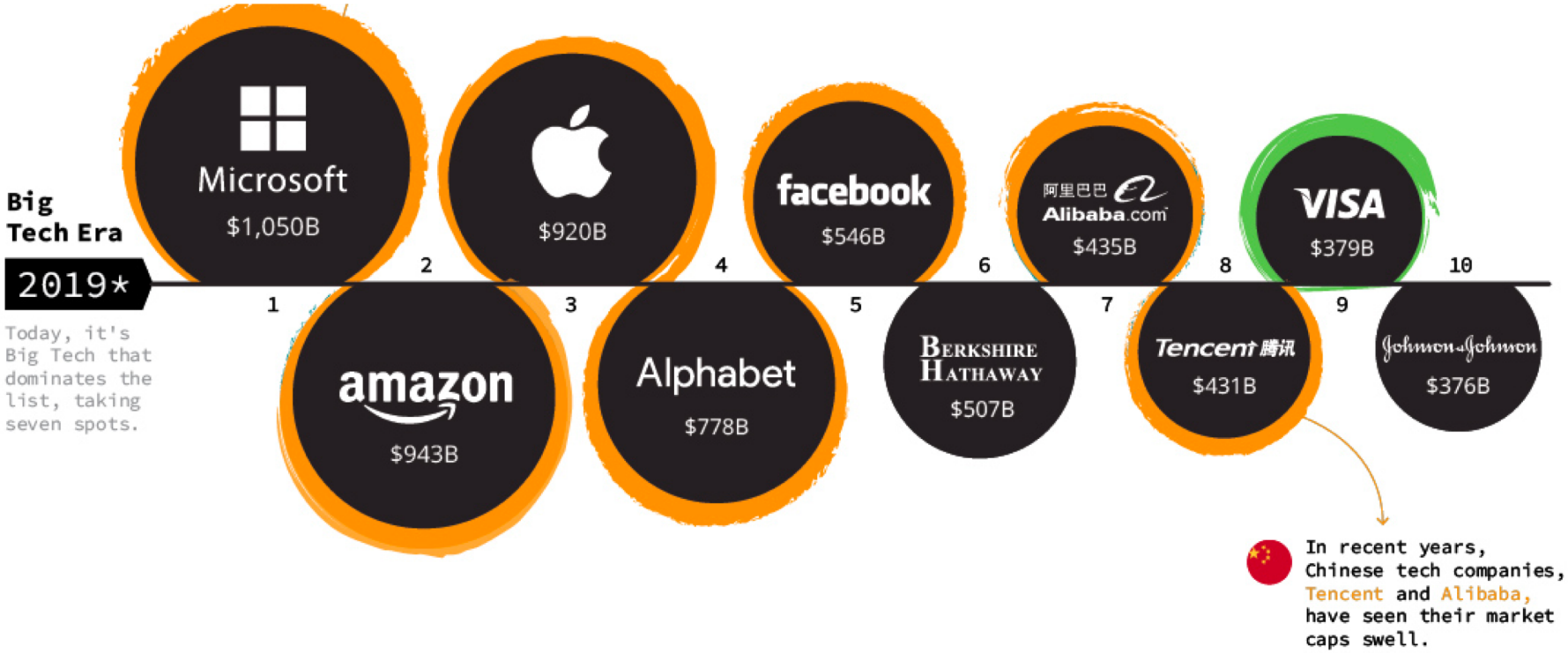
Big tech continues to dominate the world



7 of the top 10 largest companies by market cap are big tech

Source: The Visual Capitalist

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And now, big tech wants to be your bank

	Payment Applications	Credit	Money Market Funds / Insurance	# of Users	Primary Market
Alibaba.com	AliPay	X	X	~900M	China
Google	GooglePay	X		~25M	Global
Tencent 腾讯	WePay	X	X	~200M	China
amazon	AmazonPay	X		~310M	Global
	Facebook Pay Libra			~1200M	Global
	Apple Pay Apple Credit Card			~580M	Global
vodafone	M-Pesa	X		~33M	India, East Africa
mercado libre	Mercado Pago	X		~27M	South America

Big tech added payment / banking services to their platforms in an effort to tap into new revenue opportunities

Source: Statista

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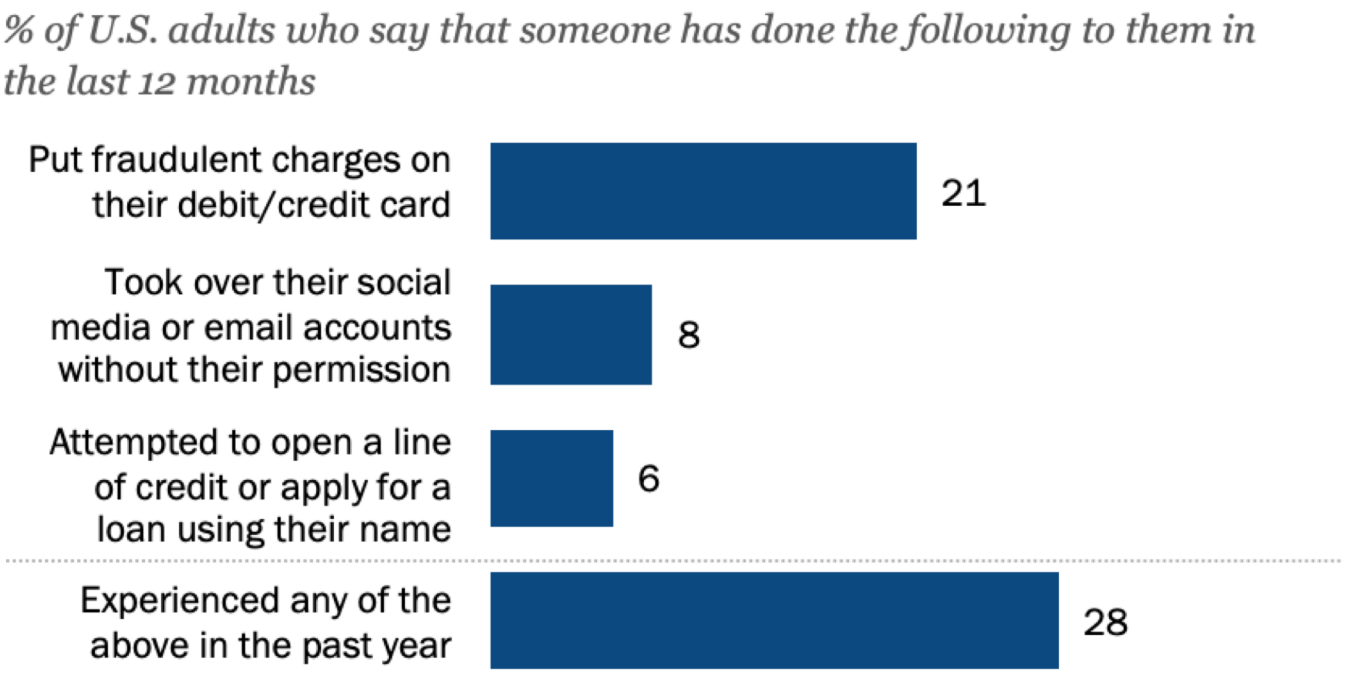
~28% of U.S. adults had fraudulent financial activity happen to them recently



As digital worlds become a large component of individual identity, stakes are higher with financial privacy than ever before

Source: Pew Research Center "Americans and Privacy" 2019

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2

Crypto Assets

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Crypto asset markets partially recovered in 2019 but remain subdued

Crypto asset market

- As a whole, crypto surged from market cycle lows but has yet to reach new highs

Bitcoin

- Bitcoin remains the market leader, with investors positioning for the May 2020 halving; innovation up the stack continues

Ethereum

- Ethereum shows strong fundamental growth despite suppressed prices; upcoming ETH 2.0 transition will be a marquis event

New chain launches

- New smart contract chain launches disappoint, with a dozen on the horizon for 2020+

Notes: All data as of 11/30/19

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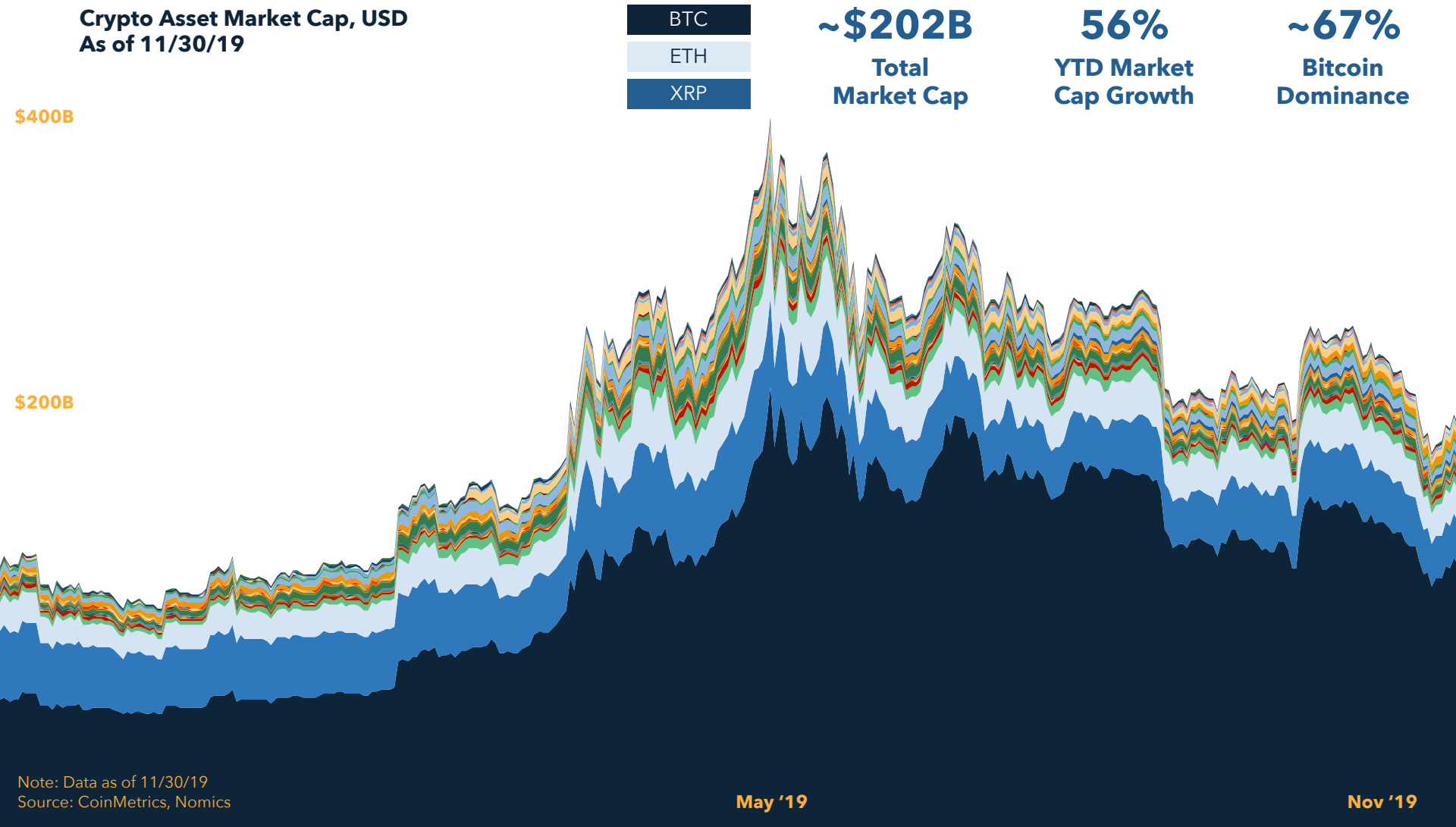
Crypto asset markets recovered in 2019, largely due to BTC and ETH

Crypto Asset Market Cap, USD
As of 11/30/19



Note: Data as of 11/30/19
Source: CoinMetrics, Nomics

Crypto markets rose ~$73B from 1/1/19 to 11/30/19, of which bitcoin accounted for 93% of growth

Top 10 Crypto Asset Prices, USD as of 11/30/19

	1/1/19	11/30/19	% Change
Bitcoin	$3,808	$7,556	98%
Ethereum	$139	$152	9%
Ripple	$0.36	$0.22	(39%)
Bitcoin Cash	$163	$218	34%
Litecoin	$32	$47	47%
EOS	$2.63	$2.75	5%
Binance	$6.19	$15.77	155%
Bitcoin SV	$91	$107	18%
Stellar	$0.11	$.06	(45%)

- Top 10 assets rallied in 2019, however, bitcoin pulled away from the rest of the market
- Binance token, BNB, led the market in terms of performance, ending the year up 154%
- Combined market cap of the top ten assets, as of 11/30/19, comprises ~76% of the total crypto asset market, compared to ~70% at year end 2018. This highlights the slight shift away from small and mid-cap tokens in 2019

Notes: Graph and table excludes Tether from Top 10, Dates as of 11/30/19
Source: CoinMetrics, onChainFX, Nomics

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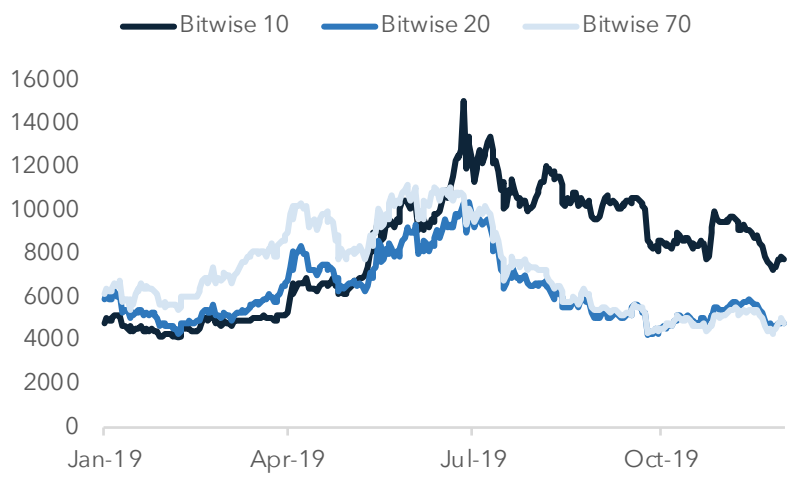
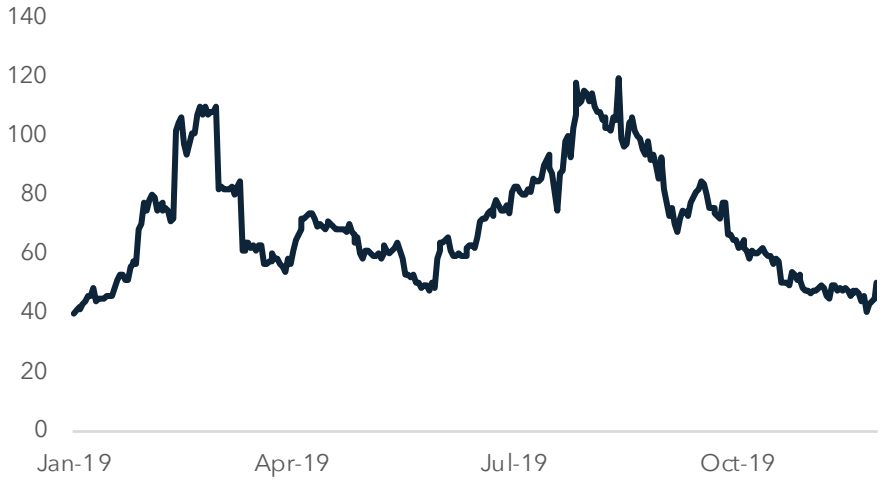
Small-mid cap index performance highlights bearish market trend

Bitwise Indices, as of 11/30/19



Bletchley Ethereum Token Index, as of 11/30/19



Commentary

- Bitwise Large Cap 10 outperformed the mid- and small-cap indices, highlighting bitcoin's market share growth over small- and mid-cap assets.
- Bletchley Ethereum Token Index had strong performance towards the end of 1Q19 and 3Q19, but overall performance flatlined by the end of the year.

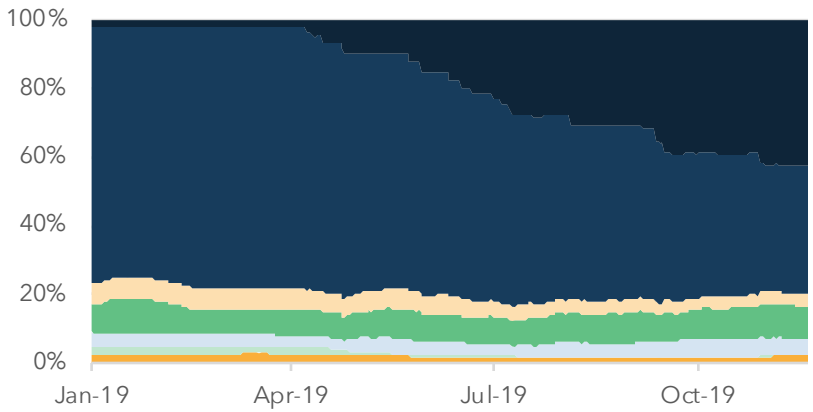
NOTES: Performance of an index is not illustrative of any particular investment. It is not possible to invest directly in an index. Performance prior to the inception date of each Bitwise Index represents a hypothetical, back-tested, and unaudited return-stream that does not represent the returns of an actual account. Index performance does not include the fees and expenses that are charged by the Fund. Actual returns may differ materially from hypothetical, back-tested returns. Visit: https://www.bitwiseinvestments.com/indexes/methodology for full methodology.

ERC-20 asset market performance uses Bletchley Ethereum Token Index with the following methodology: The Bletchley Ethereum Token Index is a market cap weighted index composed of a variable number of assets that are run on the ethereum platform. The market capitalization of each member is calculated off of the expected total supply in 2050 (pulled from OnChainFX.com) for each asset, then adjusted to reflect the publicly current available amount (float) of each asset available today (also from OnChainFX.com). There are no limits set on the weight of any individual constituent. Ethereum itself, along with ethereum classic, are not included in this index. It is composed of every token that is a member of the Bletchley Universe and is classified as ERC20.

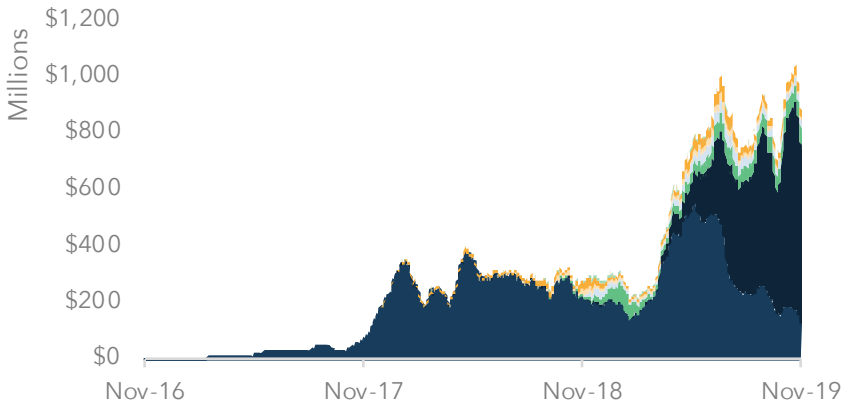
Source: CoinMetrics, Bletchley Indices, Bitwise Investments

Tether continues to dominate the stablecoin market

Stablecoin Market Dominance, as of 11/30/19



Stablecoin Transaction Value, as of 11/30/19



DAI GUSD PAXOS USDC TUSD USDT USDT_ETH

Commentary

- The total stablecoin market cap grew ~42% in 2019 from $3.3B to $4.7B, with Tether dominance at ~80%
- Unhindered by controversy over reserves and a NYAG investigation, Tether continued to dominate the stablecoin market in 2019
    - In 2019, the majority of new Tether issuance shifted away from Bitcoin and onto Ethereum (USDT_ETH) – which exacerbated congestion on the Ethereum blockchain
- Excluding Tether, majority of the remaining growth came from USDC (+76% market cap) and Paxos (+50% market cap)

Source: CoinMetrics, onChainFX

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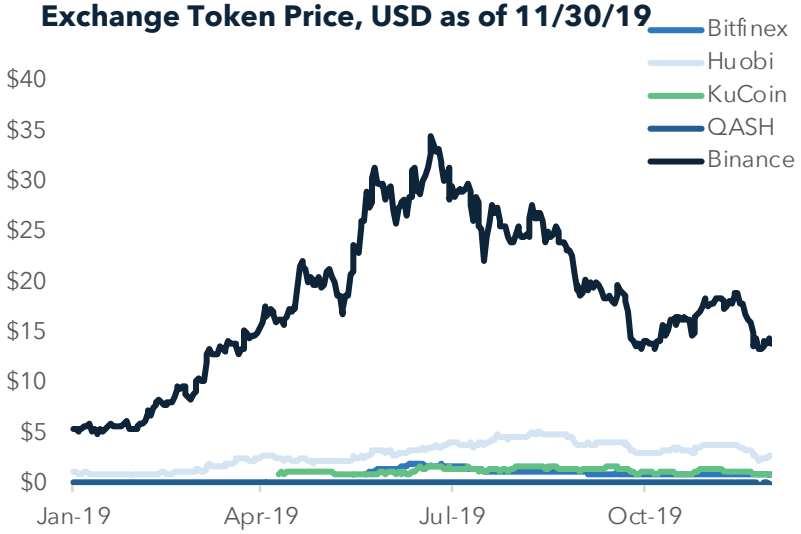
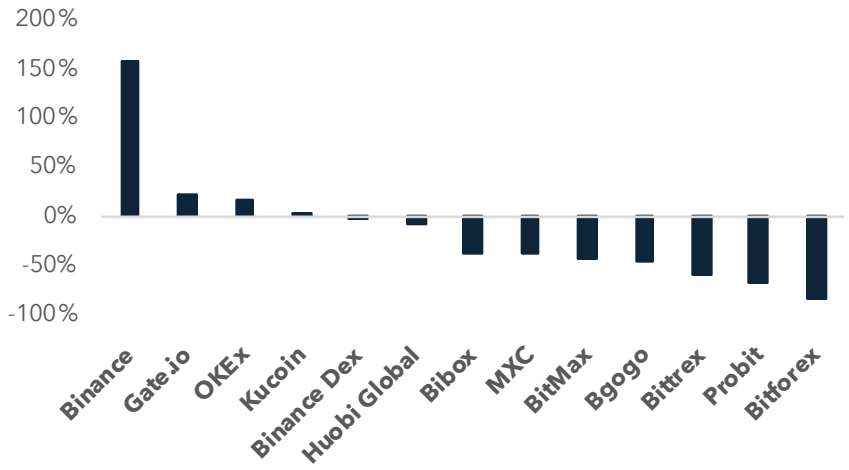
2019 saw the rise (& fall) of exchange tokens and IEOs with Binance as the major outlier

Exchange Token Price, USD as of 11/30/19



Average IEO Returns YTD by Platform



Commentary

- 2019 saw the rise of exchange tokens: tokens issued by crypto exchanges that incentivize participants to use their platform; some potential benefits included discounted trading fees, market making rewards and token burn mechanisms
- Binance pioneered the exchange token and IEO model and continues to set the pace in terms of token model innovation
- Initial exchange offerings (IEOs) emerged as a way for projects to raise capital through exchanges themselves with a direct exchange listing
- Aside from Binance, the majority of IEOs exchange platforms struggled to perform post-launch

Notes: Average returns analysis includes 95 IEOs
Source: onChainFX, CryptoRank.io

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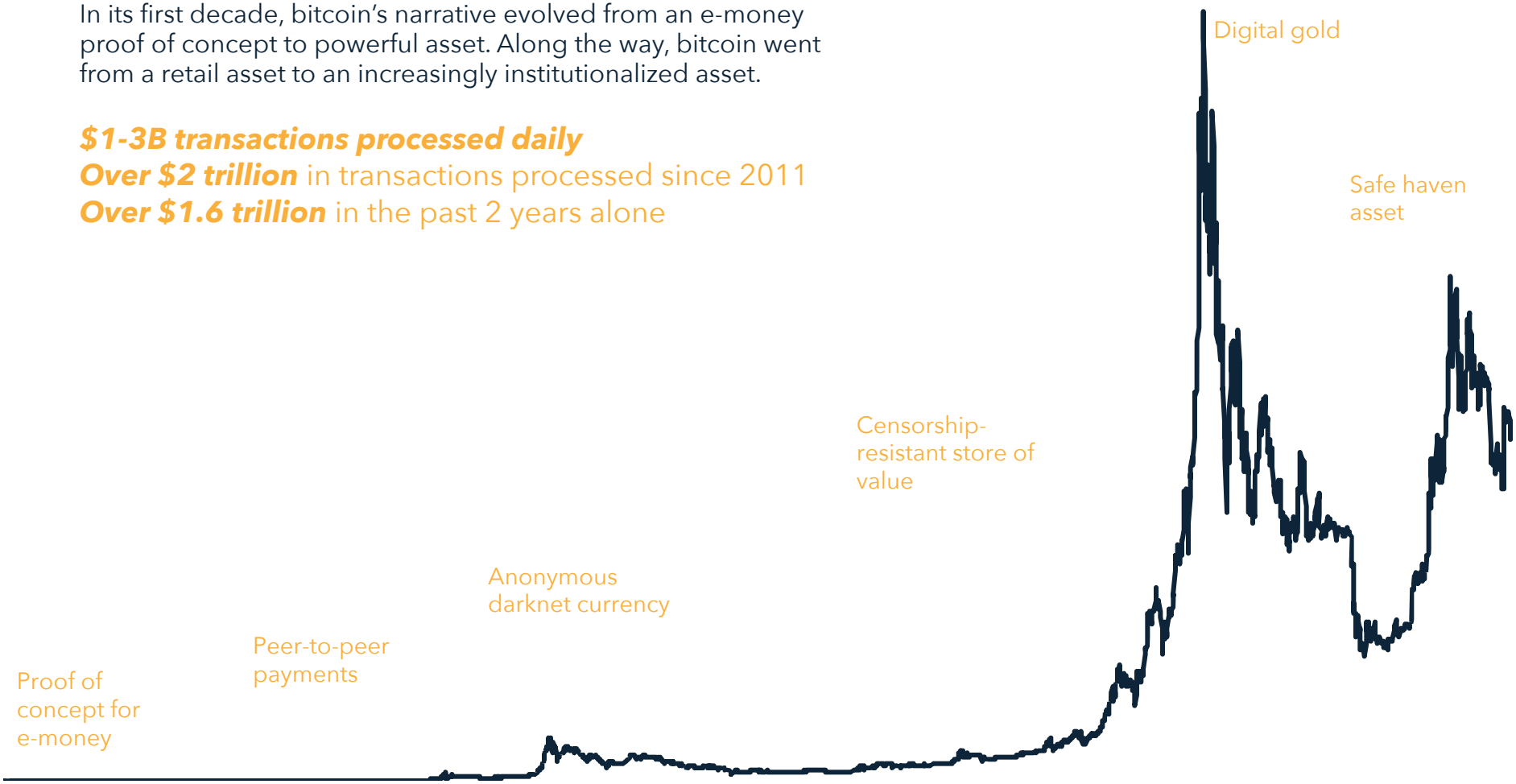
The industry converges upon bitcoin

A decade of bitcoin

In its first decade, bitcoin's narrative evolved from an e-money proof of concept to powerful asset. Along the way, bitcoin went from a retail asset to an increasingly institutionalized asset.

$1-3B transactions processed daily
Over $2 trillion in transactions processed since 2011
Over $1.6 trillion in the past 2 years alone



Source: CoinMetrics, "Visions of Bitcoin" by Nic Carter and Hasufly

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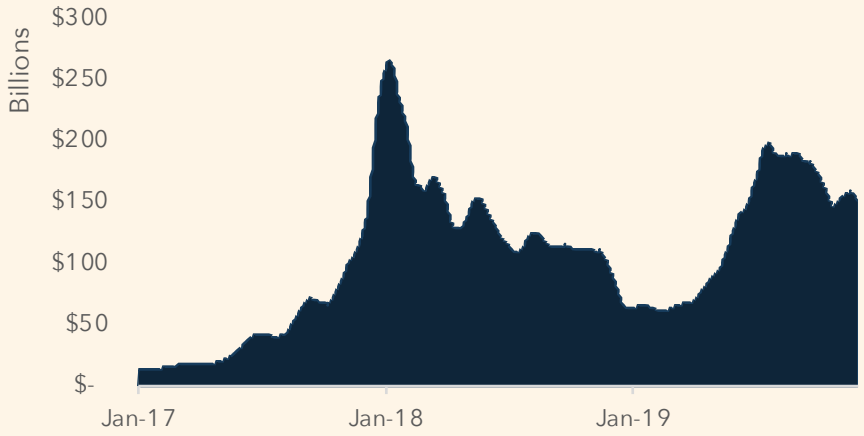
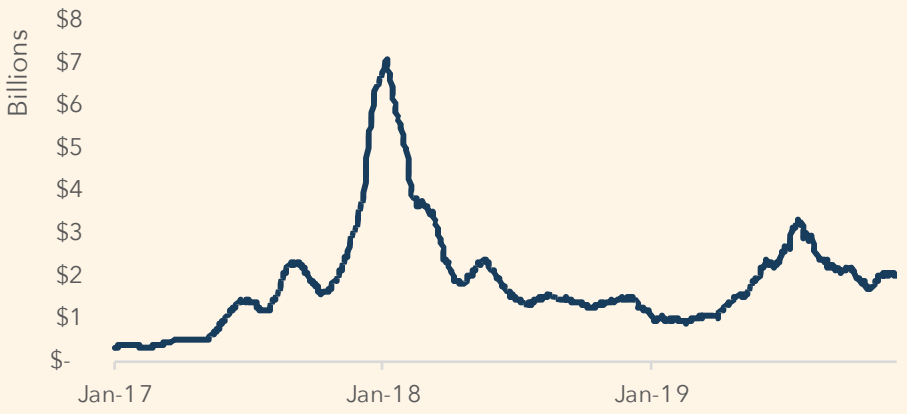
Bitcoin continues to have strong fundamentals through 2019

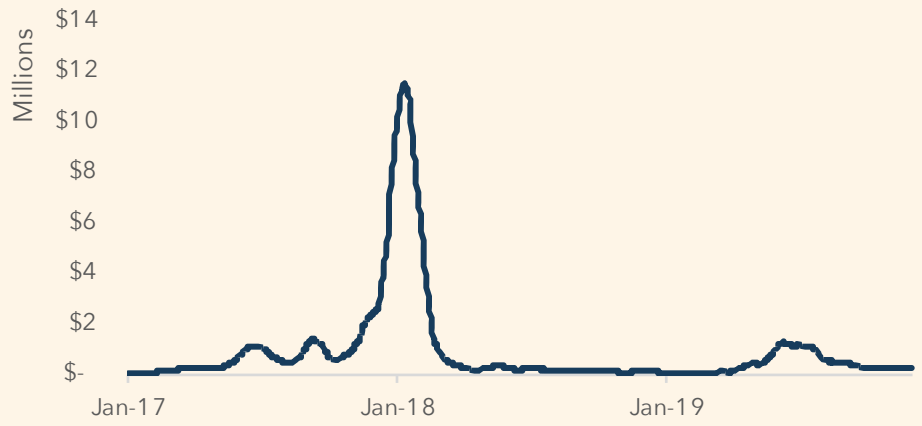
Bitcoin Market Cap, 30d Trailing Avg $ USD



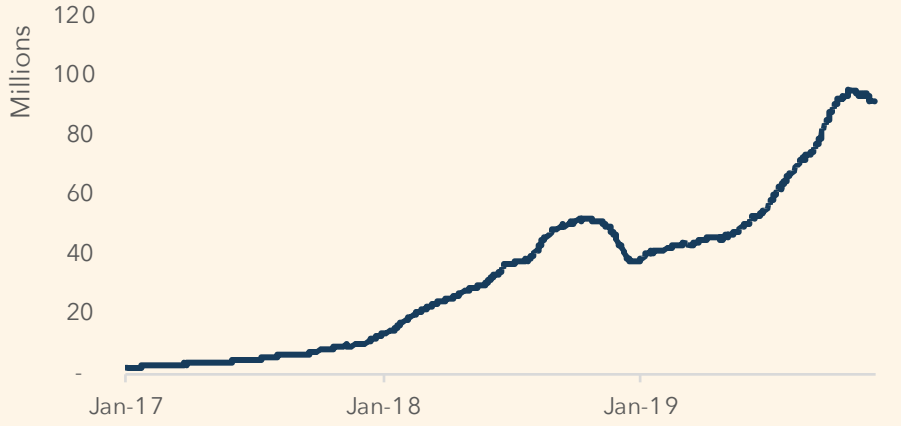
Bitcoin Adj Transaction Volume, 30d Trailing Avg $ USD



Total Fees, 30d Trailing Avg $ USD



Bitcoin Hash Rate, 30d Trailing Avg TH/s



Notes: All data as of 11/30/19
Source: CoinMetrics

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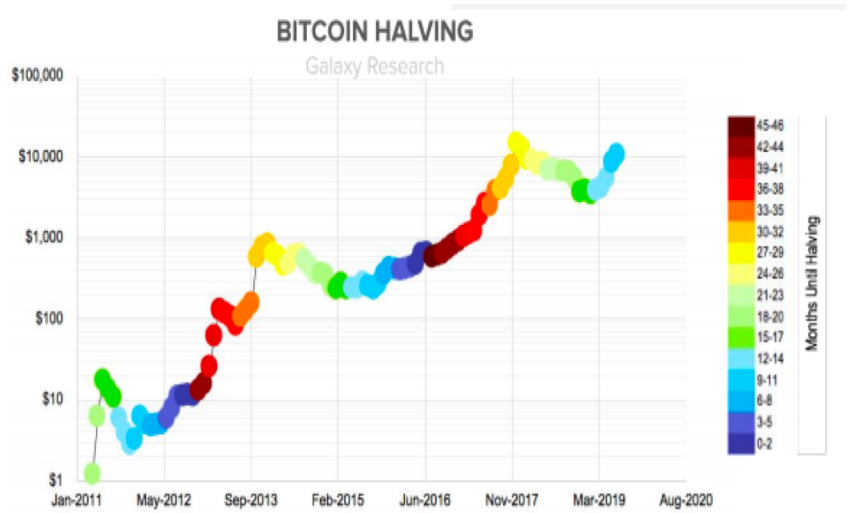
Bitcoin's third halving set for spring 2020

Dates	Block Reward	Bitcoin Issuance	Historical Price on Halving Day
2009-12	50 BTC	10.5M	$12.35
2012-16	25 BTC	5.25M	$650.63
2016-20	12.5 BTC	2.625M	--
2020-24	6.25 BTC	1.312M	--



Commentary

- Bitcoin's third halving will occur in May 2020, cutting the block reward in half to 6.25 BTC
- Bitcoin has a total supply of 21M with more than 98% of bitcoins mined by 2030, with ~86% mined as of 2019
- Although the halving is well known in advance and theoretically should be priced in, previous halvings ostensibly had significant impact on price
    - Between the first and second halving the price increased ~55x
    - Between the second and upcoming third halving, the price has increased ~12x (as of Q4 2019)

Source: Bitcoinblockhalf.com, bitcoinclock.com, Coin Metrics, Galaxy Research

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Primary focus for bitcoin continues to be privacy and scalability

	Description	Implications
Schnorr Signatures	Schnorr signatures are digital signatures that are easy to compute, highly secure, and allow for aggregation of multi-sig transactions as a normal single signature.	Schnorr signature aggregation will result in scalability enhancements, resulting in an estimated 13%[1] increase in network capacity. Additionally, privacy on Bitcoin is increased as multi-sig transactions become indistinguishable from transactions with a single signer.
Taproot	Taproot upgrade is a public change that allows for only a single public key / signature to be published, rather than including the additional hash needed in the MAST structure.	By eliminating the additional hash, Taproot provides efficiency gains. Taproot also provides privacy enhancements by hiding unexecuted conditions and making cooperative transactions look the same to a third party observer.
Liquid	Liquid is a sidechain network developed by Blockstream that offers faster and more private transaction settlement.	Liquid provides a venue for exchanges and traders to settle large amounts of bitcoin in a quicker, cheaper and more private manner. Additionally, new assets and behaviors can be enabled through Liquid.
Lightning Network	Lightning is a layer 2 solution using payment channels to create a microtransaction network layer on bitcoin.	Lightning enables cheaper, faster and more private payments on top of Bitcoin, opening up new use cases such as micropayments.

Notes: (1) Based on BitMex Schnorr Signatures research
Source: CoinMetrics, BitMex Research

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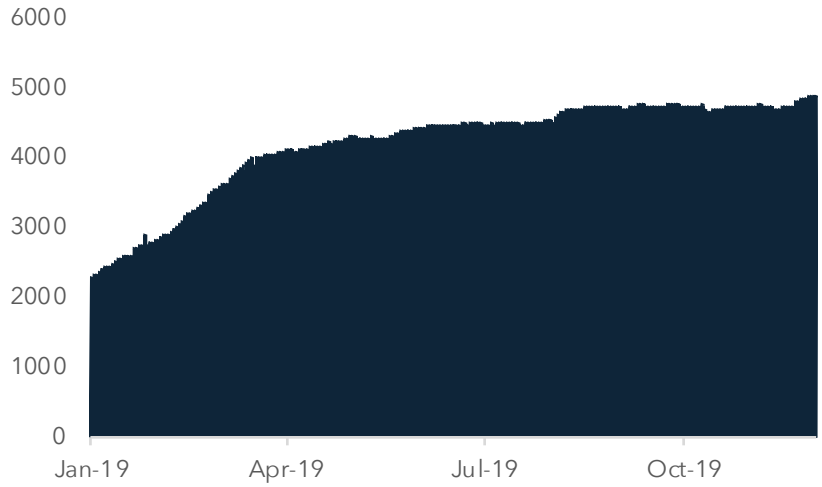
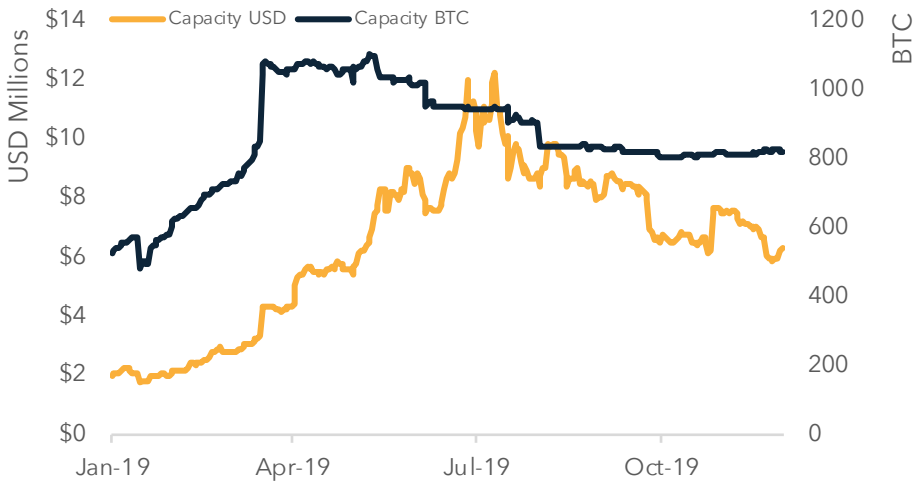
Lightning network channels and capacity demonstrate slow but steady growth

Nodes with Channels, as of 11/30/19



Lightning Network Capacity, as of 11/30/19



Commentary

- Lighting network now has 5K+ nodes, 30K channels and current network capacity is ~$6M, up 3x from Jan 2019
- Lightning Labs improved Lightning usability, launching the mobile app and releasing light client support through Neutrino
- Venture funding accelerated with new tools and applications including OpenNode, Acinq, RADAR ION and Casa. Merchant products enable payments with consumer products like BitRefill, Fold and LN.Pizza. Despite the influx of Lightning-based projects, scalable revenue-generating models remain an outstanding question

Source: Bitcoin Visuals

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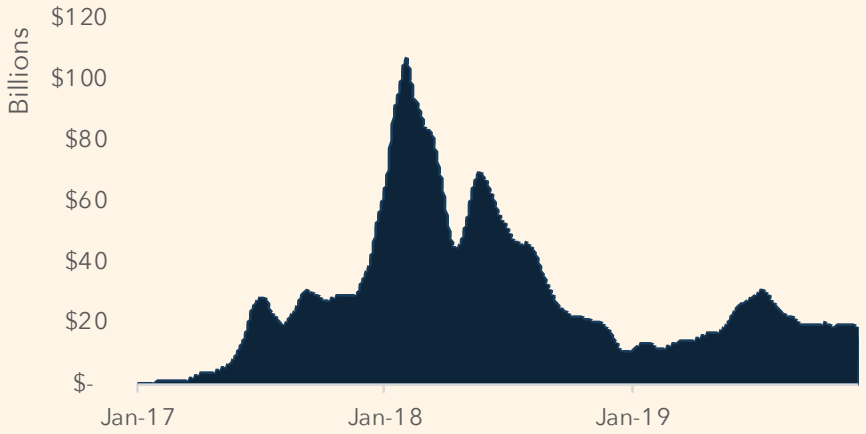
Ethereum usage grew in 2019 despite suppressed prices

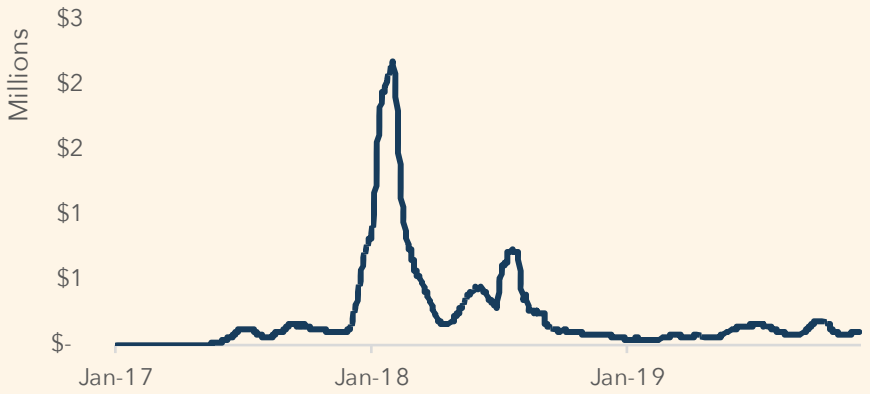
Ethereum Market Cap, 30d Trailing Avg $ USD



Ethereum Contract Calls, 30d Trailing Avg

Ethereum Active Addresses, 30d Trailing Avg

Total Fees, 30d Trailing Avg $ USD



Notes: Data as of 11/30/19
Source: CoinMetrics, Etherscan

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ETH 2.0 transition is in flux; expected to take multiple years

Estimated Planned Transition from ETH 1.0 to 2.0

Phase 0: Beacon Chain

Introduce new chain and new asset (ETH 2)

First building block of ETH 2.0

Phase 1: Shard Chains

Design and delivery of shard chains. Focus on construction, validation and consensus of shard chains

Phase 2: Execution

Smart contracts reintroduced, with support for accounts, contracts and other familiar validations

Phase 3: Light Client State Protocol

Minimized on-chain state by moving data off chain

Commentary

- The Ethereum Foundation has $30M of capital allocated towards supporting Eth 1.0 ($8M), Eth 2.0 research and launch ($19M) and growing the developer ecosystem ($3M)
- Ethereum's planned transition to Eth 2.0 is expected to rollout over the next 3-5 years, with initial launch of the beacon chain in Q1 of 2020
- The transition management strategy is in flux; likely that Eth 1.0 and Eth 2.0 will exist in parallel for some time
- The next planned iteration of Ethereum will remove the ability to execute transactions atomically, which may compromise the overall composability of Ethereum dApps
- There are a dozen teams working on Eth 2.0; however many look to Vitalik as a source for direction and strategy

Source: Ethereum Foundation, Vitalik Buterin Blog, DevCon

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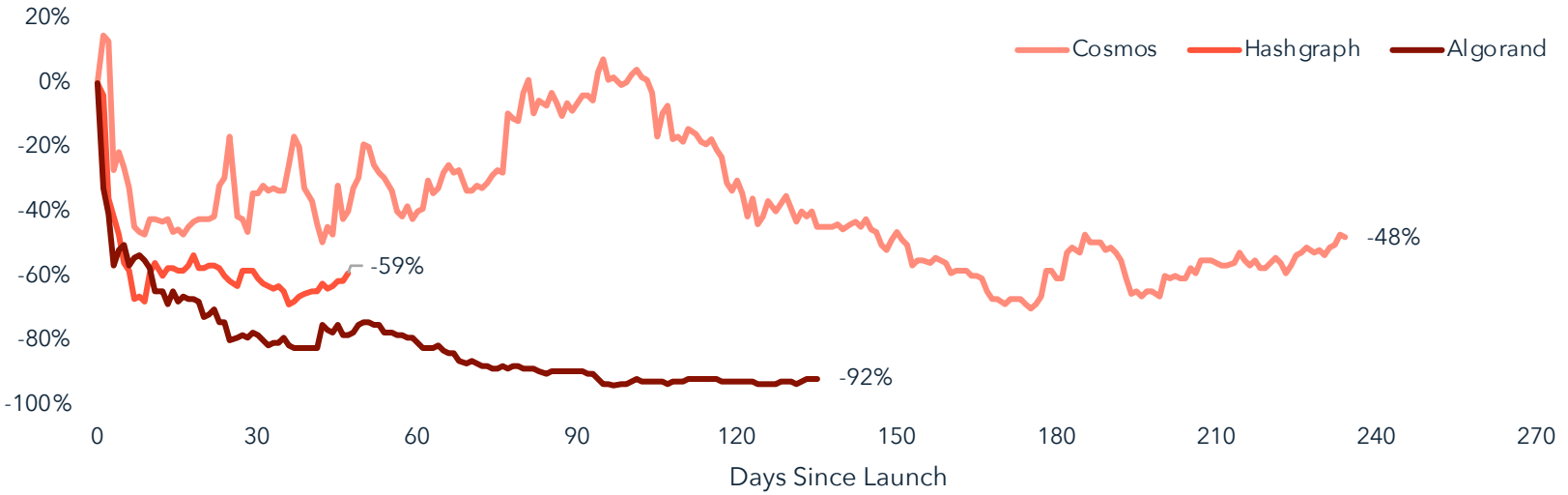
2019 crypto asset launches largely disappoint

Post-Launch Price Performance of New Assets



Commentary

- 2019 saw the launch of a few high-profile Proof-of-Stake chain launches, namely Cosmos ($17M+ raised) Hashgraph ($100M+ raised) and Algorand ($60M+ raised)
- Each of these chains struggled from a price perspective as they failed to attract significant developer or user attention
- Most of these projects have launched 'ecosystem funds' in an attempt to catalyze development activity and, ultimately, user demand. However, past ecosystem funding efforts have tended to only be successful in recruiting "mercenaries" collecting easy funding rather than "missionaries" with longer-term conviction

Source: CoinMetrics

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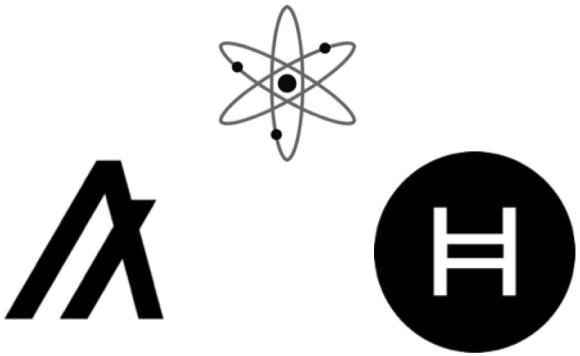
“ETH killer” chain hasn’t emerged;15+ new launches upcoming

2019 Launches



Upcoming Launches

Commentary

- Over the course of 2019, Bitcoin and Ethereum maintained ~70% market cap dominance of the entire crypto asset market cap, despite a few high-profile launches
- In the next 12 months, billions of dollars of VC capital will be unlocked and available for sale in the market as 15+ protocols go live
- In the search for the next smart contract platform, ethereum and bitcoin are likely immediate beneficiaries of these launches as a portion of capital shifts from new launches back to these assets

Source: Various project websites

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3

Adoption

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Varying adoption across market segments

Retail Users

- Bitcoin continues to resonate with millennials
- Full time crypto protocol developers grew in 2019

Institutional Interest

- Traditional institutions and big tech firms revealed large crypto projects
- Sophisticated infrastructure develops further in the crypto ecosystem

Nation States

- Facebook's Libra awakened governments globally to digital currencies
- China's PBoC plans for a digital currency came under spotlight

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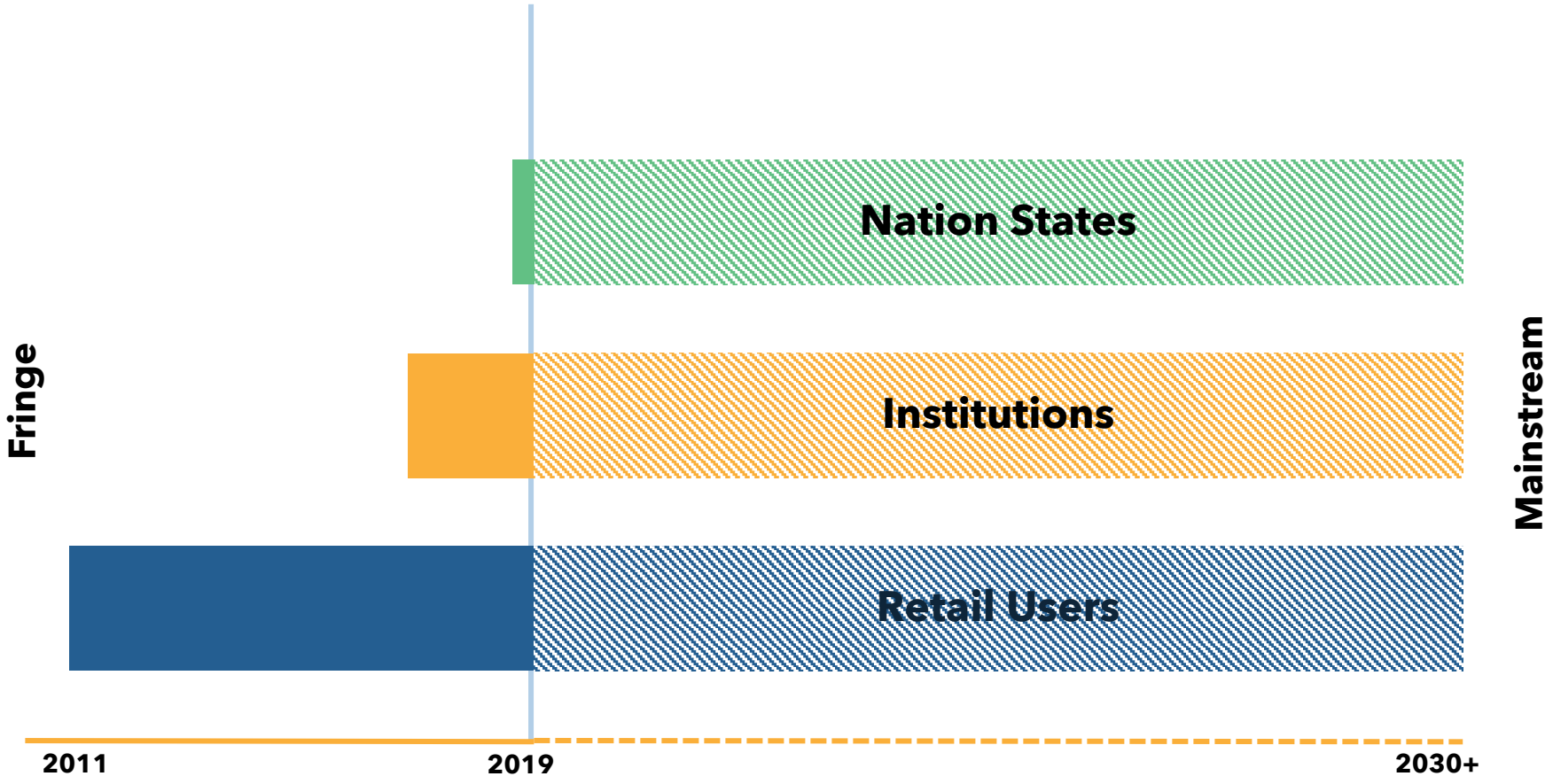
Crypto assets are going from fringe to mainstream across three major user segments



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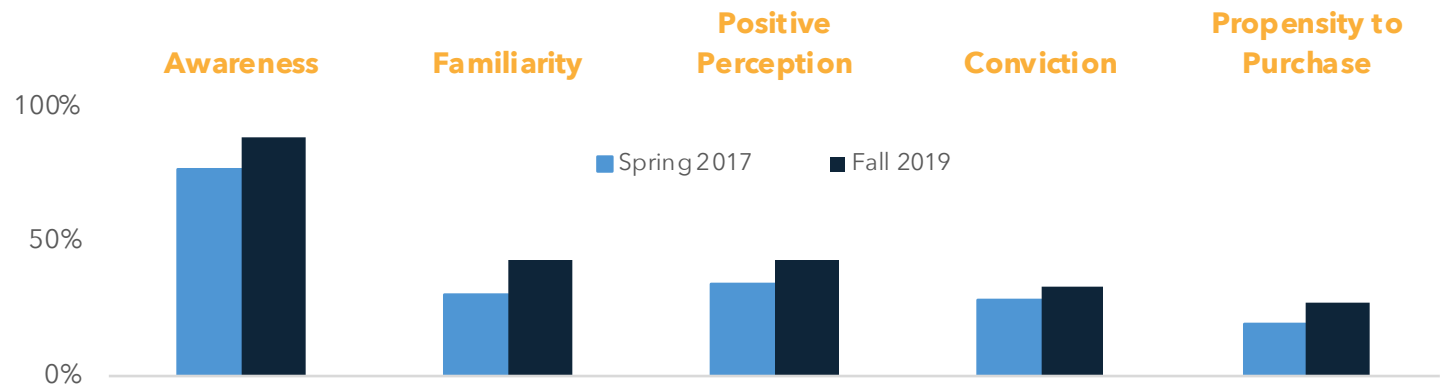
Bitcoin is a demographic mega-trend, resonating with millennials and Gen-Z

Bitcoin sentiment in following categories…



Bitcoin resonates particularly with millennials and Gen Z (ages 18-34)

~50% strongly/somewhat agree that <u>most people</u> will be using Bitcoin in the next 10 years

Notes: This survey was conducted online within the United States between April 23-25, 2019 among 2029 adults (aged 18 and over) by The Harris Poll on behalf of Blockchain Capital via its Harris On Demand omnibus product. Figures for age, sex, race/ethnicity, education, region and household income were weighted where necessary to bring them into line with their actual proportions in the population. Propensity score weighting was used to adjust for respondents' propensity to be online.
Source: Blockchain Capital Harris Poll Survey

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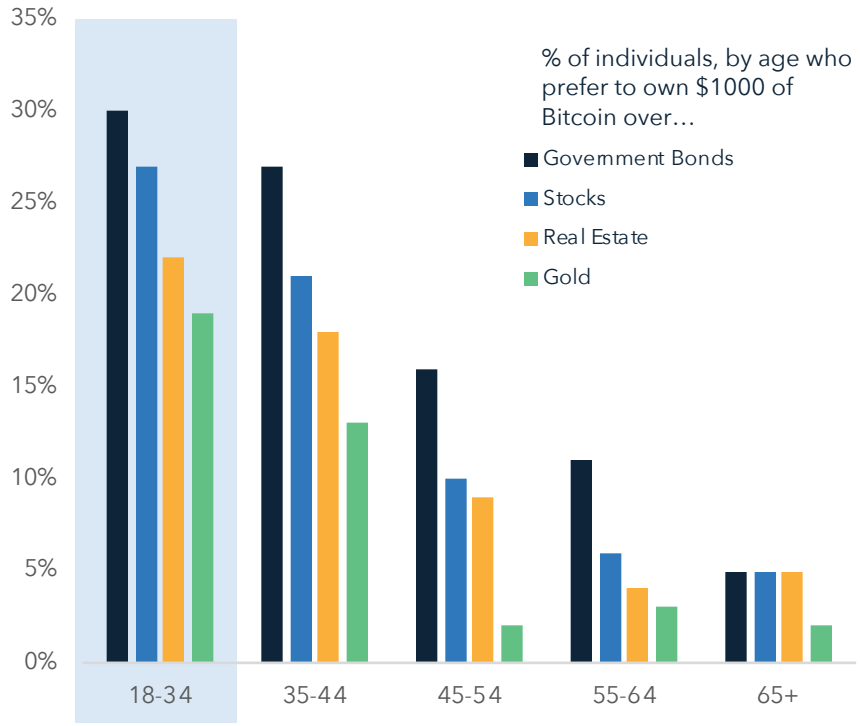
As generational wealth transfer begins, investment preferences may be a bitcoin catalyst

An estimated $15T of wealth is expected to transfer by 2030 to younger generations…

…making their investment preferences increasingly meaningful



Source: Wealth-X Consulting Firm, Blockchain Capital Harris Poll Survey

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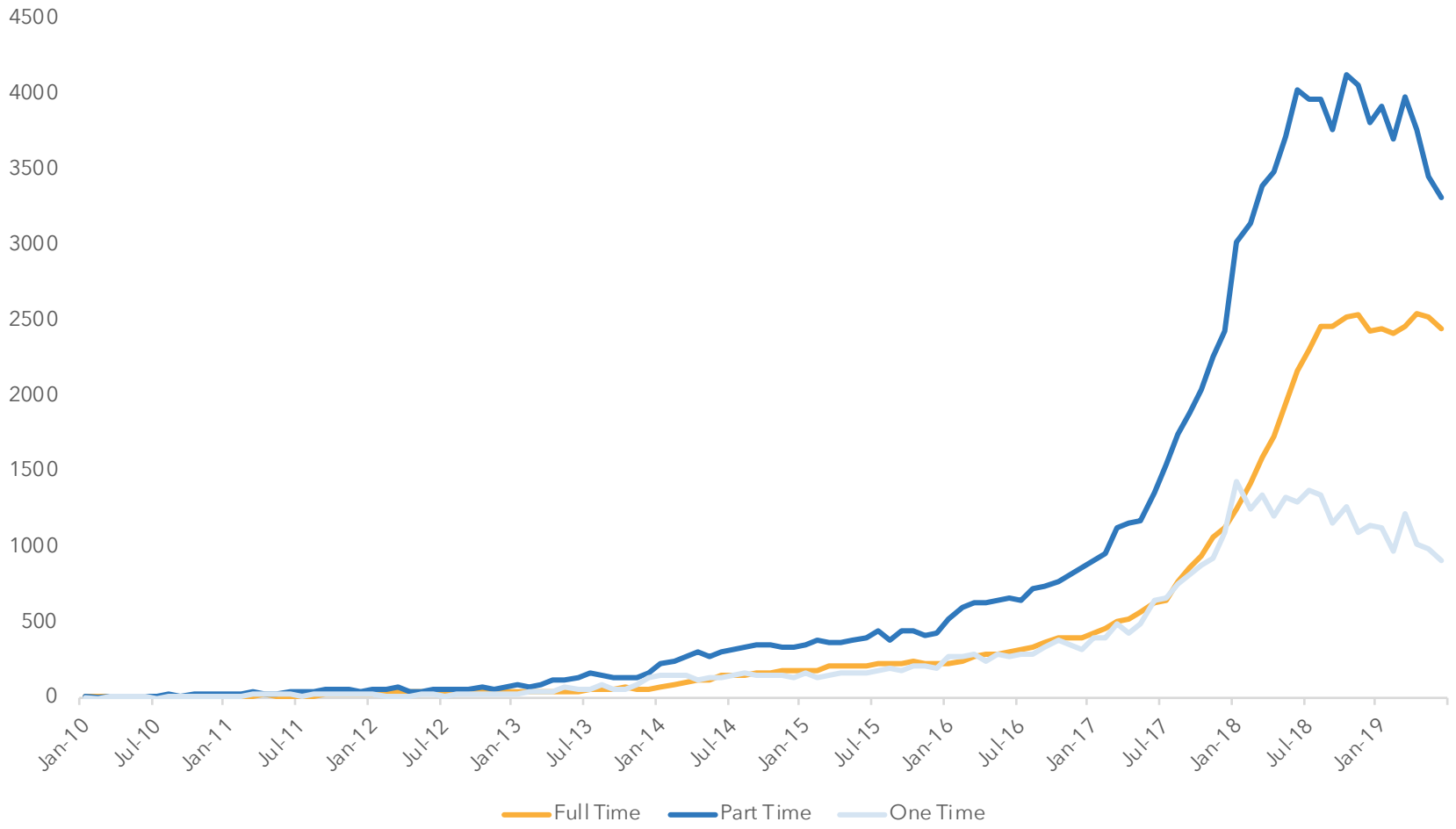
Full time blockchain developers increased 13% from June ‘18 to June ‘19



Source: Electric Capital 1H19 Developer Report

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Industry made leaps in terms of end user experience and access to accelerate adoption

Improved UI / UX

New wallets and products are creating intuitive experiences, obfuscating confusing user steps like address and gas fees

argent

UNSTOPPABLE DOMAINS

Increased access

Crypto payment processors are focused on increasing access with frictionless fiat onramps such as ApplePay

wyre

MoonPay™

Exposure to new demographics

Startups targeting new demographics seek to tap into familiar ecommerce patterns (e.g., cashback rewards)

gracias

lolli

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Financial institutions make major moves as crypto financial products mature

Traditional FIs entering the space

- **Fidelity Digital Assets** rolled out their custody offering
- **TD Ameritrade** offers clients ability to invest in bitcoin futures via ErisX
- **JPMorgan** announced their enterprise stablecoin for B2B transactions
- **Bakkt** (ICE/NYSE owned) launches their bitcoin futures to the market

Crypto FIs mature in the market

- **Anchorage, Gemini and Coinbase** become institutional qualified custodians
- **BlockFi** competes in the traditional bitcoin lending space
- **Routefire, Tagomi, Talos Trading** entered the space to provide liquidity aggregation and tooling

Commentary

- Five years ago, most FIs were dismissive of bitcoin and blockchain – today it feels imprudent to not have a blockchain strategy. In many ways bitcoin is increasingly viewed as a reputationally de-risked asset
- FI distribution channels have potential to enable access for tens of millions of new bitcoin users, as well as mainstream adoption for the entire asset class (both retail and institutional clients)
- Despite nascent institutional support for bitcoin, there is massive room for growth among retail and institutional segments alike; education and awareness are necessary ingredients

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Fintech doubles down on crypto products to meet demographic demand

Square

$28B Mkt Cap

Cash app functionality to buy / sell bitcoin, as well as send to family members

Square sold more than $125M of bitcoin in 2Q19 alone

robinhood

$5.6B Mkt Cap

Robinhood offers commission free crypto trading for some US states

Crypto assets are not transferrable in and out of Robinhood custody right now

SoFi

$1B+ Mkt Cap

SoFi offers crypto trading with SoFi invest in some states

Currently offers trading for bitcoin, ethereum and litecoin; it is a closed system with assets remaining in SoFi

eToro

$800M+ Mkt Cap

eToro offers trading for 14 assets

eToro also launched a cryptocurrency trading strategy for users based on social media sentiment regarding various assets

Commentary

- Given the demographic overlap, Fintech is extremely well positioned to enable crypto on-ramps and capture demand for crypto trading
- Square emerged as a leading platform, bringing on high profile bitcoin talent to build bitcoin products and funding open source development
- As more fintech players get involved, opportunity to grow the crypto retail user base expands significantly

Source: Square financials, Yahoo Finance

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Social media giants look to launch their own blockchains and tokens

Geography	Users	Project	Response
Global	~2.4B	Stablecoin backed by basket of currencies	US regulators reacted extremely negatively - questioning the intentions of Facebook and potential harm of Libra on the U.S. dollar
		Calibra wallet for supporting products	Global leaders expressed uncertainty, instigating conversations around digital currencies at the central bank level.
		Libra Association to govern Libra token	Zuckerberg claims the project will not launch until regulators approve.
Global	~200M	Telegram raised $1.7B to launch in-app currency TON	The launch of TON has been halted by an SEC emergency action in Fall 2019
Korea	~50M	Klaytn, Kakao's blockchain, went live in 2019 with a slew of social and gaming apps	Klaytn launched with 10+ dapps and has been growing users ever since
Japan	~217M	LINE, a leading Southeast Asia messenger, launches their own crypto exchange	Project went live Sept 2019 for 80 million users in Japan with full regulatory approval

Source: Statista, Libra, Calibra, Coindesk, Blockchain Capital

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The separation of money and state was tested by Libra and China's digital currency plans

China's Cryptocurrency Plan Has a Powerful Partner: Big Brother

Facebook's Libra project led Beijing to accelerate its efforts. The government could soon know a lot more about how people are spending.

CHINA BUSINESS & ECONOMY

China Readies State-Controlled Digital Currency

'Digital yuan' poses huge challenges to privacy and Western banking hegemony

BY FAN YU, EPOCH TIMES
September 22, 2019 Updated: September 24, 2019

Share

China's digital currency will kick off 'horse race': central bank official

Various sovereign central bank digital currency efforts rumored to be underway

Source: Wall Street Journal, Epoch Times, Reuters, The Block

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Central bank digital currencies heavily explored by a number of nations

What is a CBDC?

A CBDC is a new form of legal tender, different from other types of money that central banks currently issue (i.e., cash, reserves). Instead, CBDC is the equivalent of digital cash. While proposed approaches vary, in most, the central bank plays a critical function. At a minimum, the central bank maintains the reserves backing the digital cash. In more involved approaches, the central bank may play a role in issuing the digital cash, as well as creating and supporting the underpinning technology.

Why create a CBDC?

First, a CBDC offers the benefits touted by digital payments broadly. As a means of payment, CBDC could increase liquidity and scalability for domestic transactions. Digital cash also makes it easier to export and drive adoption of domestic currency abroad – which potentially augments any nation's influence on the global economy and international politics

However, some central banks may issue CBDC to surveil their citizens' digital activities and be able to trace financial transactions more efficiently. This intent would be destructive to financial freedom and privacy.

What efforts are underway?

In 2019, rumors emerged that Japan, France, the EU, Ghana and Tunisia, in addition to China are all evaluating or planning a digital currency. The U.S. is also said to be evaluating a digital dollar and its implications.

Remains unclear if CBDCs will function like cash or a bank account, but each model has widely different implications

Source: IMF, The Block, CoinDesk, Committee on Payments and Market infrastructure

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4

Regulation

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Regulatory landscape remains a complex and critical area for crypto assets

Token issuance & distribution

- SEC announced settlements or actions for major projects like EOS, Sia and Telegram
- Overall, regulation of token issuance remains unclear

Money transmission & B/D licensing

- FINCEN provided clarity around the requirements for money transmitters
- SEC awarded a B/D license and two Transfer Agent licenses

Global outlook

- Regulators increasingly addressed crypto assets in particular, clarifying guidance by proposing or adopting specific laws

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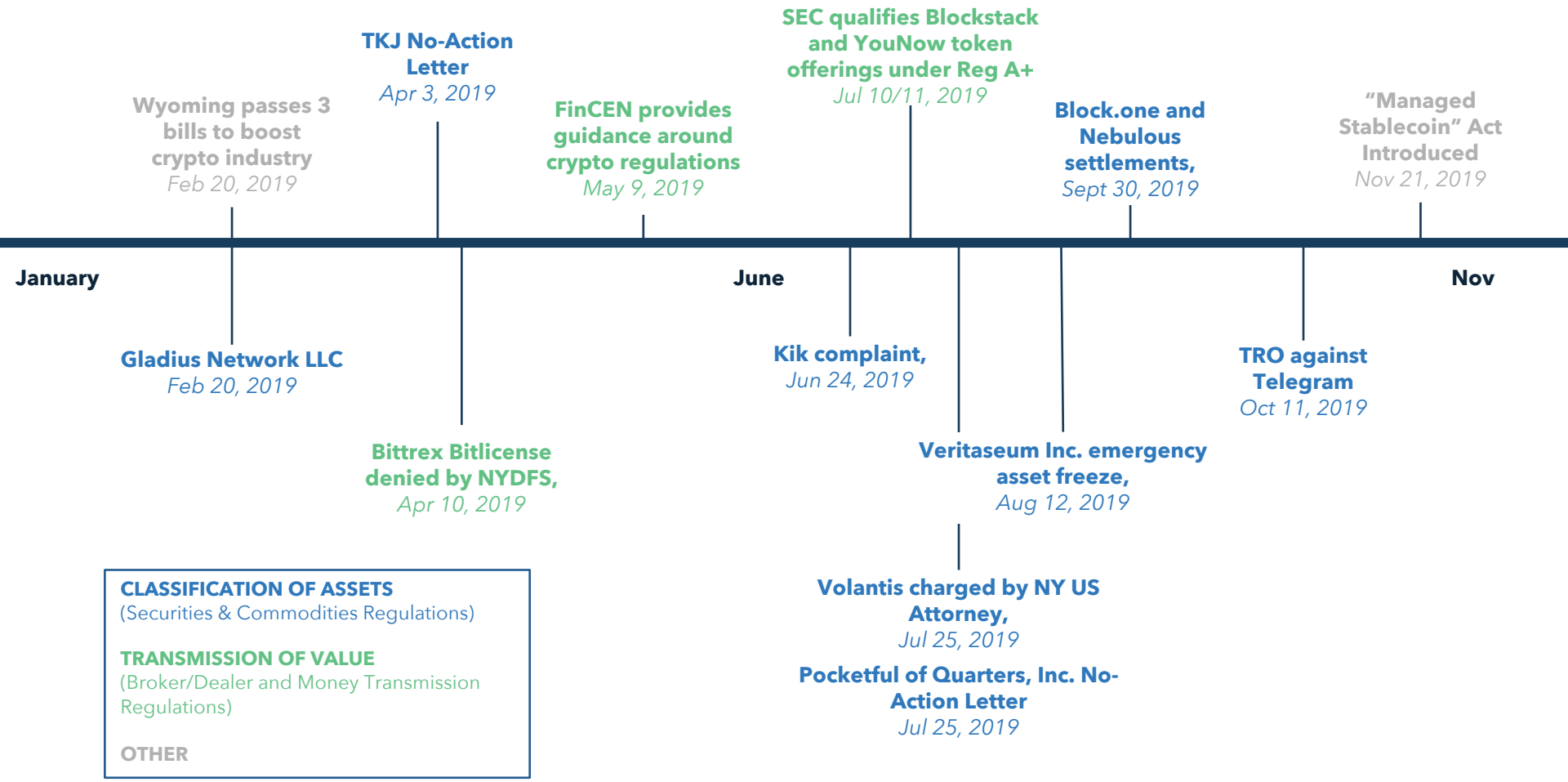
Two regulatory areas emerged in the U.S.: classification of assets and transmission of value



Source: FinCEN, SEC, Blockchain Capital Analysis

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Token issuance continues to be an uncertain regulatory area

First, SEC suggests certain ICOs violated securities laws…

Projects were penalized and forced to register tokens as securities and / or rescind tokens entirely

…then EOS, a $4B token sale, was deemed an unregistered offer of securities, with a different outcome…

EOS was penalized, with no registration or rescission required, seeming to indicate the SEC deemed EOS tokens to NOT be securities

…finally, SEC filed an emergency restraining order against Telegram

Telegram sold to institutional investors and never planned a public token sale. Their token distribution was blocked by the SEC restraining order

It is likely that there are true standards for decentralization and/or demonstrated use case, but each case differs by the "facts and circumstances"

Source: Blockchain Capital Analysis

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FinCEN provided guidance around money transmitters

FinCEN provides guidance offering a framework for classifying money transmitters for business models involving "convertible virtual currencies." The guidance suggests the following types of designations:

NOT Money Transmitters:

- Non-custodial wallets
- Decentralized exchanges
- "*software*" providers
- dApp developers

Money Transmitters:

- Custodial wallets
- Custodial exchanges
- "*service*" providers utilizing dApps to transmit value

BisonTrails

Ledger

TRM

BINANCE

coinbase

Source: FinCEN guidance

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Positive regulatory clarity around money transmission and regulatory approvals

Regulatory Approval	Implications
Blockstack and YouNow tokens approved for a Reg A+ qualified offering	Provides guarantees there is a legally compliant path for token funding and distribution
Eight projects received their NY Bitlicense approval, including: Tagomi, Bitstamp, LibertyX, SoFi and Robinhood	Notoriously stringent, the NY Bitlicense approval means projects can expand their offerings in a key market
Securitize approved for first ever digital asset transfer agent	SEC recognizes a transfer agent as an important part of the capital market ecosystem for crypto assets
First non-custodial digital asset broker-dealer license approved for Harbor, with others next in line	FINRA approval may suggest a path forward for digitally native broker - dealers, after a long wait-and-hold period for most applications

Source: Blockstack, Securitize, Harbor

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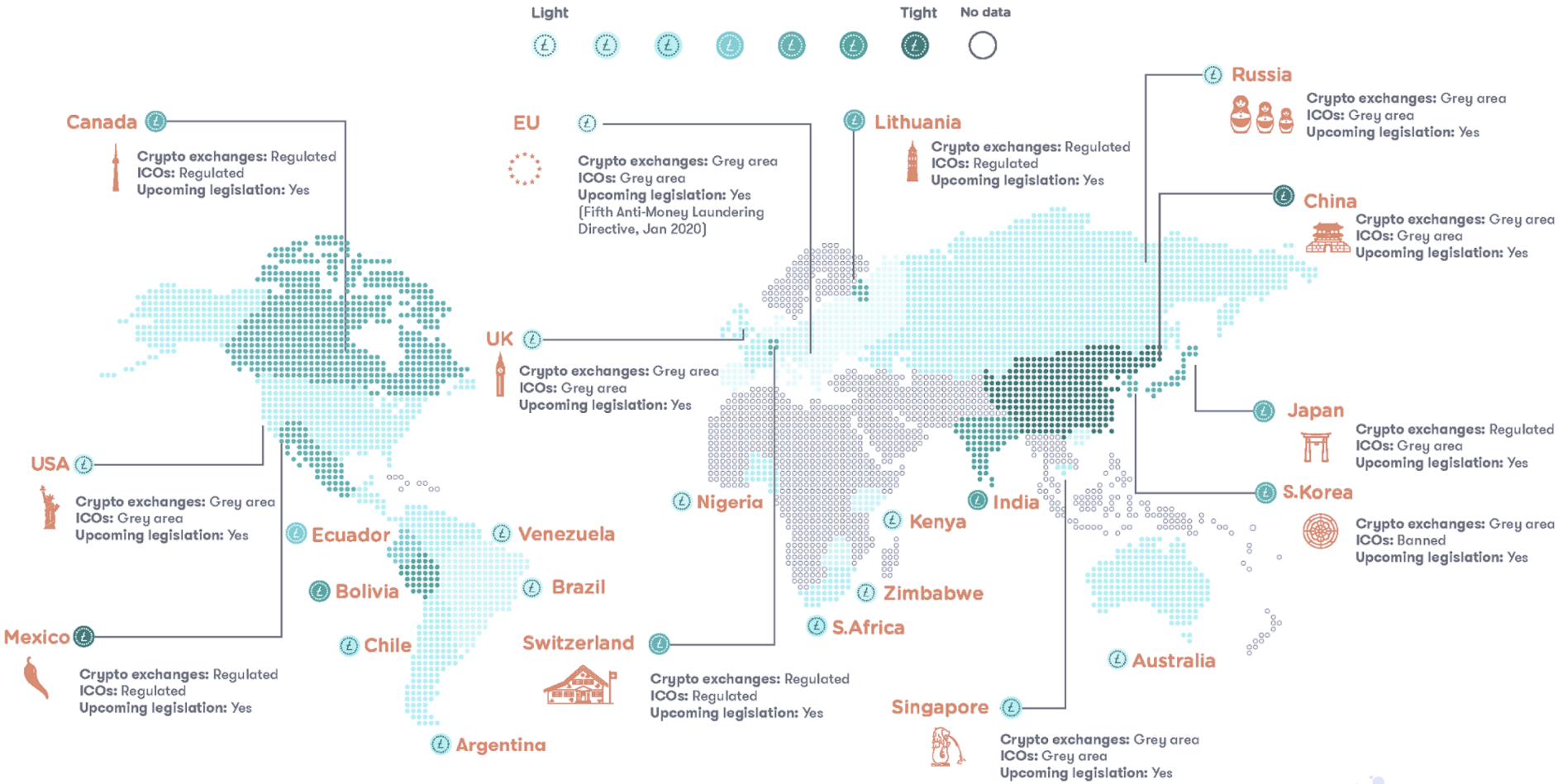
Crypto assets are regulated at a global level



The light-to-tight regulation scale is based on the following criteria: are Cryptocurrency Exchanges and ICOs banned, regulated or operating in a grey area? Ban = 3 points, regulated = 2 points, grey area = 1 point. Legal Tender? Yes = 1 point, No = 0 points. Is there any planned legislation to increase crypto regulation? Yes = 1 point, No = 0 points. Data collected by ComplyAdvantage should be used as a guide and never taken as legal advice.

ComplyAdvantage

Source: Comply Advantage

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However, regulation widely varies across geographies

	Major Events	Stance
China	President Xi announced new initiatives to position China as a blockchain leader Potential for China to loosely use blockchain based technology for a new digital yuan	Pro enterprise blockchain use cases
Japan	Japan continues to regulate crypto exchanges, recently granting a licenses to messaging giant LINE Rumors that Japan may be developing a SWIFT-like network for crypto payments	Cautious but still positive regulatory environment
Hong Kong	Hong Kong regulators determine new rules for crypto exchanges to get licensing	Cautious but still positive regulatory environment
UK	FCA proposed crypto asset classification framework to allow token issuers to create innovative tokens that are not securities	Cautious but still positive regulatory environment
EU	FATF proposed travel rule suggests KYC / AML will be set at a much higher threshold for exchanges and wallets	High degree of regulation towards the crypto industry
India	Continued ban on exchanges, harsh regulation for blockchain projects	Negative sentiment

Source: Wall Street Journal, The Block, Coindesk

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5

Major Themes

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Major themes of 2019

DeFi	▪ Financial infrastructure built using protocols and smart contracts emerged as a key use case in 2019
Prime Brokerage	▪ Prime brokerage stack continues to grow, with sophisticated crypto players offering services from custody and exchange to settlement and execution
Gaming	▪ Major protocols continue to push forward on gaming plans despite any major traction among users
Developer Tools	▪ Data, node deployment and developer tools emerged to fill a critical gap in making blockchain-based product development easier

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The DeFi ecosystem grew rapidly over the course of 2019

Source: Defi Pulse, DefiPrime

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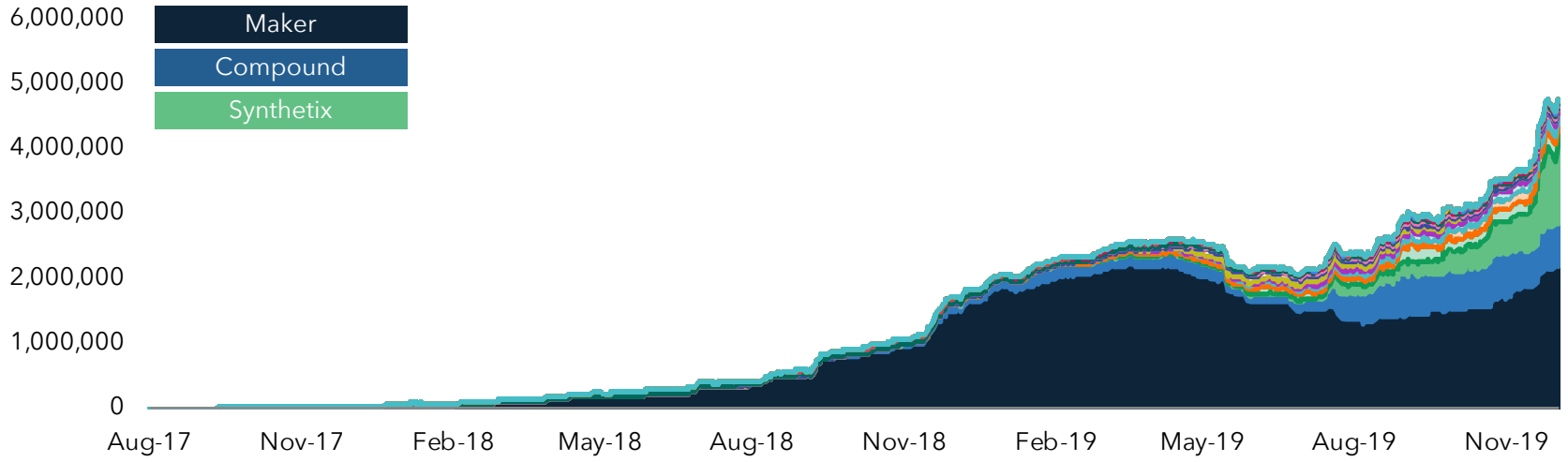
Over $600M of ETH is locked in open finance smart contracts

Total Value Locked (ETH) in "Decentralized Finance" projects



Commentary

- ~84% of value is locked in top three projects: Maker, Compound and Synthetix.
- Maker dominance declines to <50% of all ETH locked in smart contracts
  - The DAI borrowing rate spiked over 2019, reaching 20.5% in August at its peak; since then it has been declining
- Defi Pulse now tracks over 20 projects in the DeFi ecosystem, with Compound, Synthetix, and Uniswap as breakouts in 2019

Source: Defi Pulse

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Over the course of 2019, $600M+ in DeFi loans were originated

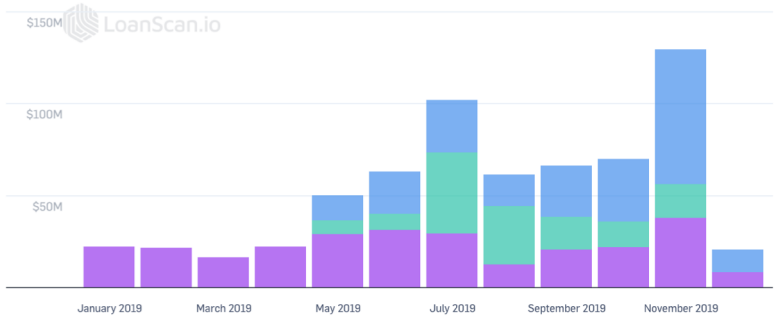
Loans Originated, USD

YTD VOLUME

$648,348,919

YTD TRANSACTIONS COUNT

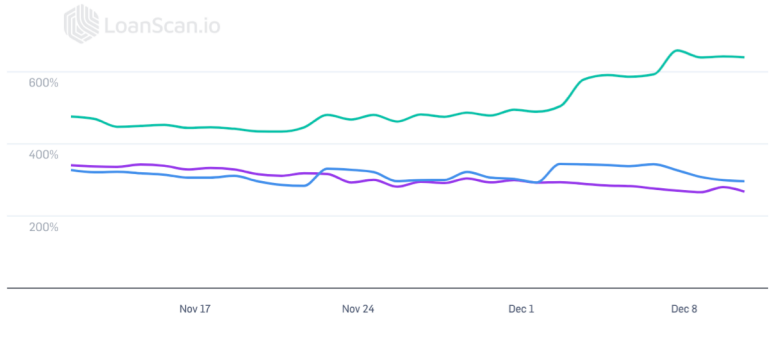
282,302



Collateral Ratio, %

WEIGHTED AVERAGE

349%



Commentary

- Across lending platforms, $648M in loans were originated in 2019; as of Jan '19, only $22M in loans had been originated, a 28x increase
- The primary motivations for borrowers were: leverage for traders and working capital for crypto-native companies
- Assets are primarily stablecoins like Dai; however the introduction of multi-collateral Dai will introduce new collateral options for loans
- Lending products today are primarily highly overcollateralized, making them less capital efficient
- Identity and reputation remain key outstanding ingredients for improving the capital efficiency of on-chain lending and expanding the addressable market

Source: Loanscan.io, Defi Pulse

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Competition for prime brokerage services grew intense

In its **current state, crypto prime brokerage services are unbundled**

Likely to see **consolidation into 2020+**, as providers move towards **vertical consolidation to offer better pricing, convenience and optionality**

Source: Company logos

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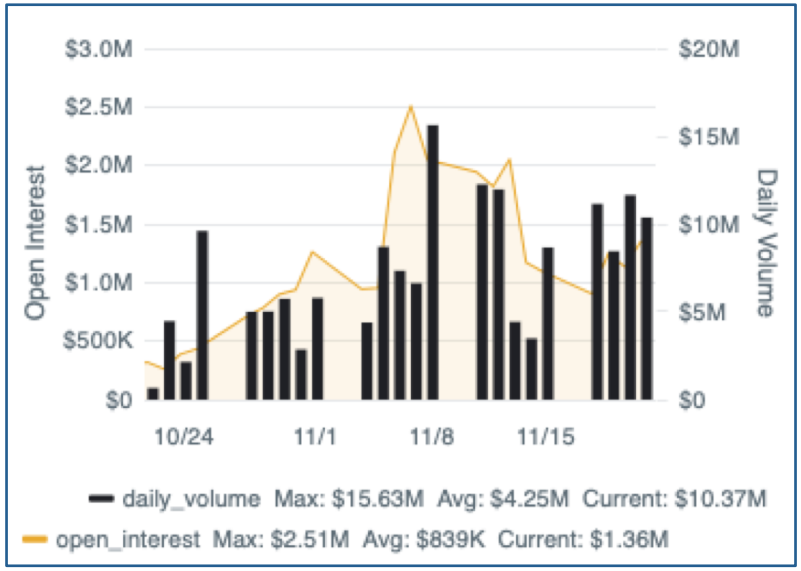
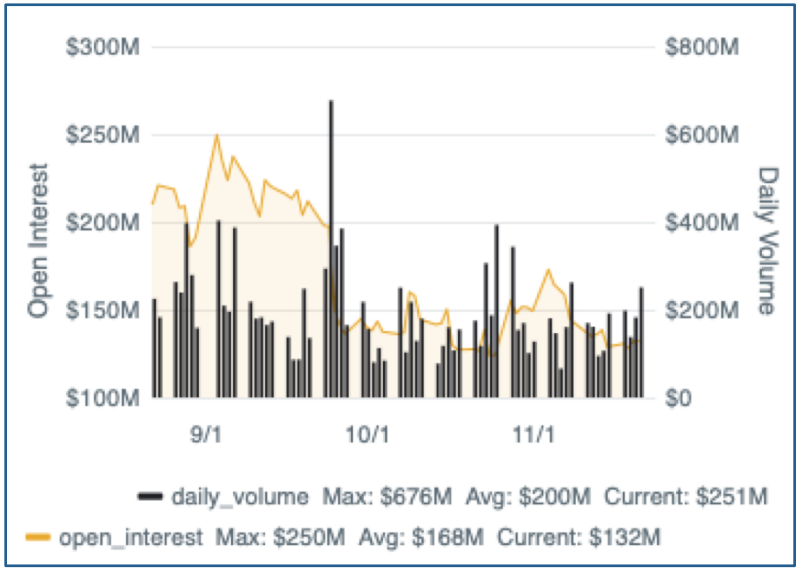
Bitcoin derivatives are improving price discovery in the market

Bakkt bitcoin futures total open interest and volume, $ USD



CME bitcoin futures total open interest and volume, $ USD



Commentary

- Bakkt bitcoin futures launched in September, reaching ~$10M daily volumes of and $1.36M in open interest by November
- CME bitcoin futures saw average daily volumes of $200M and average open interest of $168M higher than the usual
- While CME continues to dominate Bakkt, the gap is slowly decreasing as Bakkt looks to catch up

Source: Skew, Coindesk, CME, Bakkt

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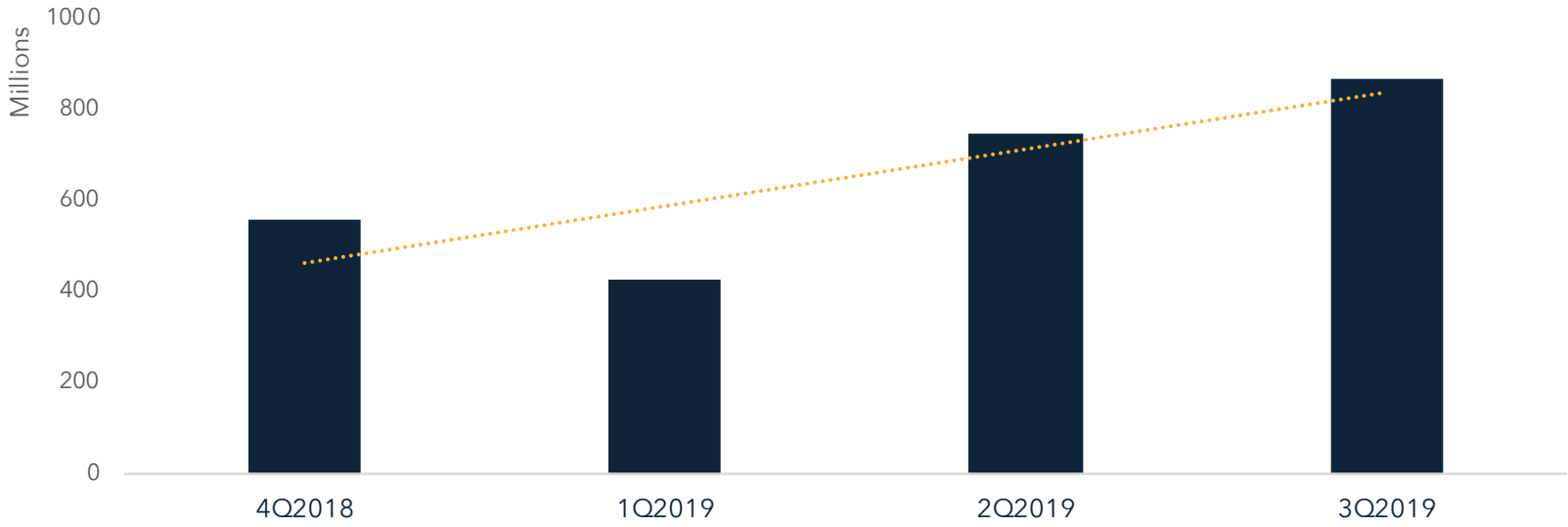
Crypto-backed lending growth indicates strong institutional interest

Genesis Capital lending originations, $ USD



Commentary

- Growth in crypto-collateralized borrowing from centralized lending desks highlights appetite among industry participants
- As the industry's largest and most liquid asset, bitcoin comprised a significant portion of Genesis Capital's lending and borrowing demand
- BTC-backed cash lending to international and Asian counterparties grew significantly for Genesis Capital, presenting an opportunity to earn attractive yields

Source: Genesis Lending Quarterly Reports

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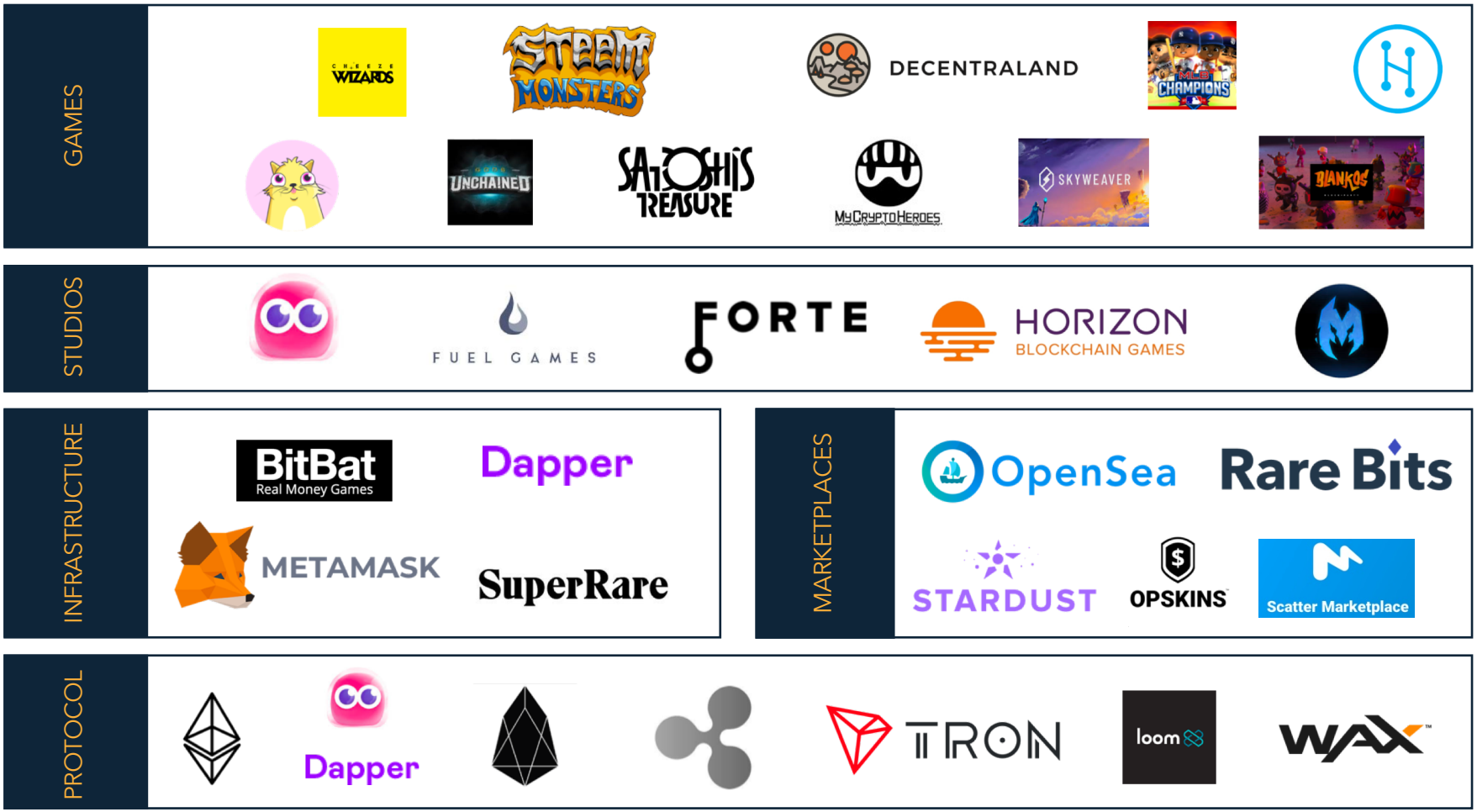
The blockchain gaming ecosystem continues to develop



Source: Various project websites

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Ethereum remains leading protocol for games, despite a flurry of competition

Protocol	Description	Games
ETHEREUM	Leading smart contract platform, creator of ERC-721 open standard	Unchained, Decentraland
FLOW (DAPPER)	CryptoKitties creator's base layer protocol for scalable blockchain-based games	CryptoKitties, Cheeze Wizards
RIPPLE (INTERLEDGER)	Payment protocol for conducting transactions across different blockchains	Forte
EOS	DPoS chain focused on higher throughput and scalability	Blankos, Hero
LOOM SIDECHAIN	Sidechain similar to Plasma, with gaming dapps as side chains	Relentless
STEEM	Social media network using DPoS	Steem Monsters

Source: Various project websites

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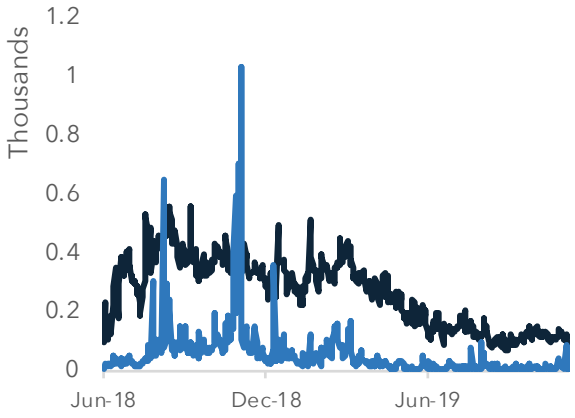
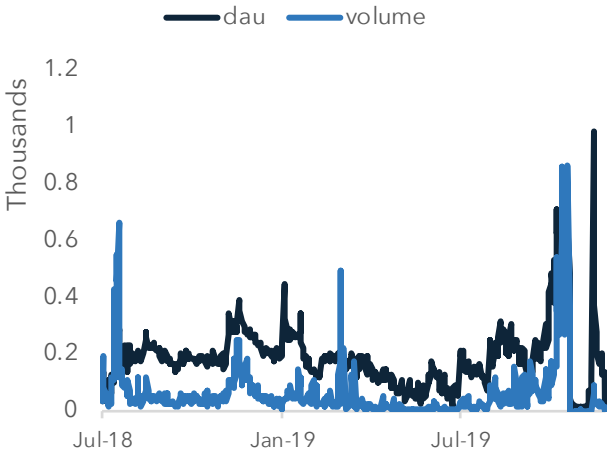
Despite buzz, blockchain gaming activity remains underwhelming with no hits

Among most popular live games, traction has flatlined or remained low

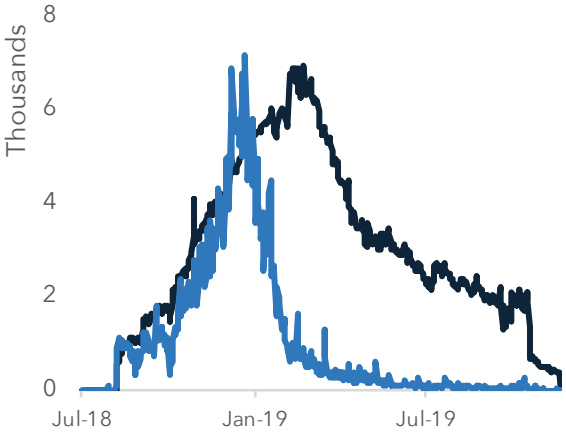
Crypto Kitties, Ethereum / Dapper



Gods Unchained, Ethereum



EOS Knights, EOS



Total usage remains low

<1K dApps across gaming / gambling

<20K Average DAU for gaming

Source: Dapp Radar, State of the Dapps

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Developer infrastructure and tools make building crypto products easier

NODE INFRASTRUCTURE

BISON TRAILS

DEPLOY

BLOCKDAEMON

INFURA

DAppNode

QuikNode

TESTING & TOOLS

TRUFFLE

Ganache

openzeppelin

Quantstamp

DATA & ANALYTICS

Digital Assets Data

COINMETRICS

amberdata

nomics

KAIKO

BLOCKCYPHER

TokenAnalyst

Etherscan

Source: Various websites

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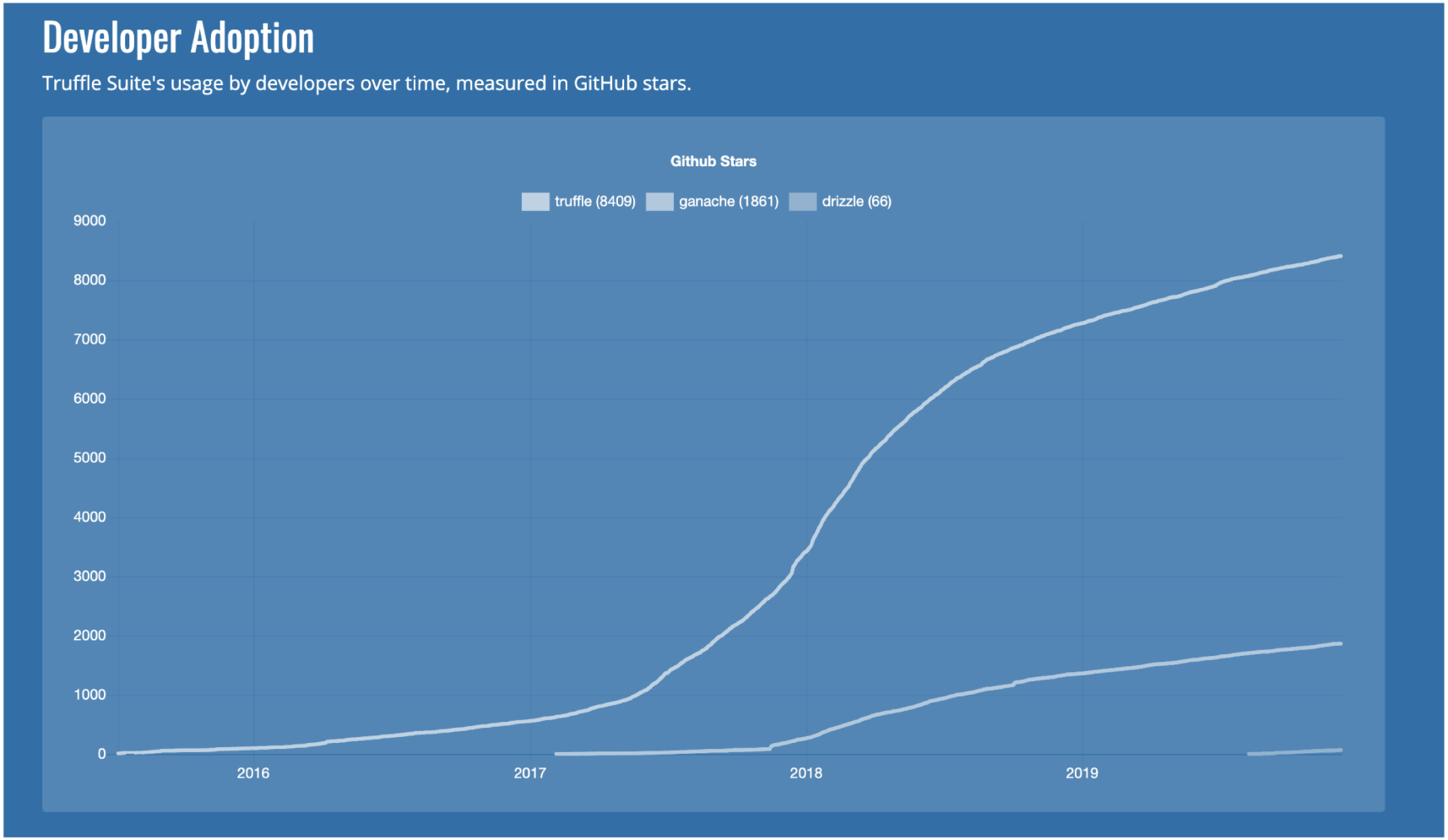
Anecdotal evidence suggests demand for developer tools appears on the rise



Source: Truffle

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Looking Ahead

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Looking ahead

Industry Maturation

- Crypto remains in early stages, with a majority of user traction congregating around onboarding (exchanges)
- As industry matures, expect increase in M&A activity

Predictions

- Our team shares our bold predictions around the regulatory landscape, DeFi and more

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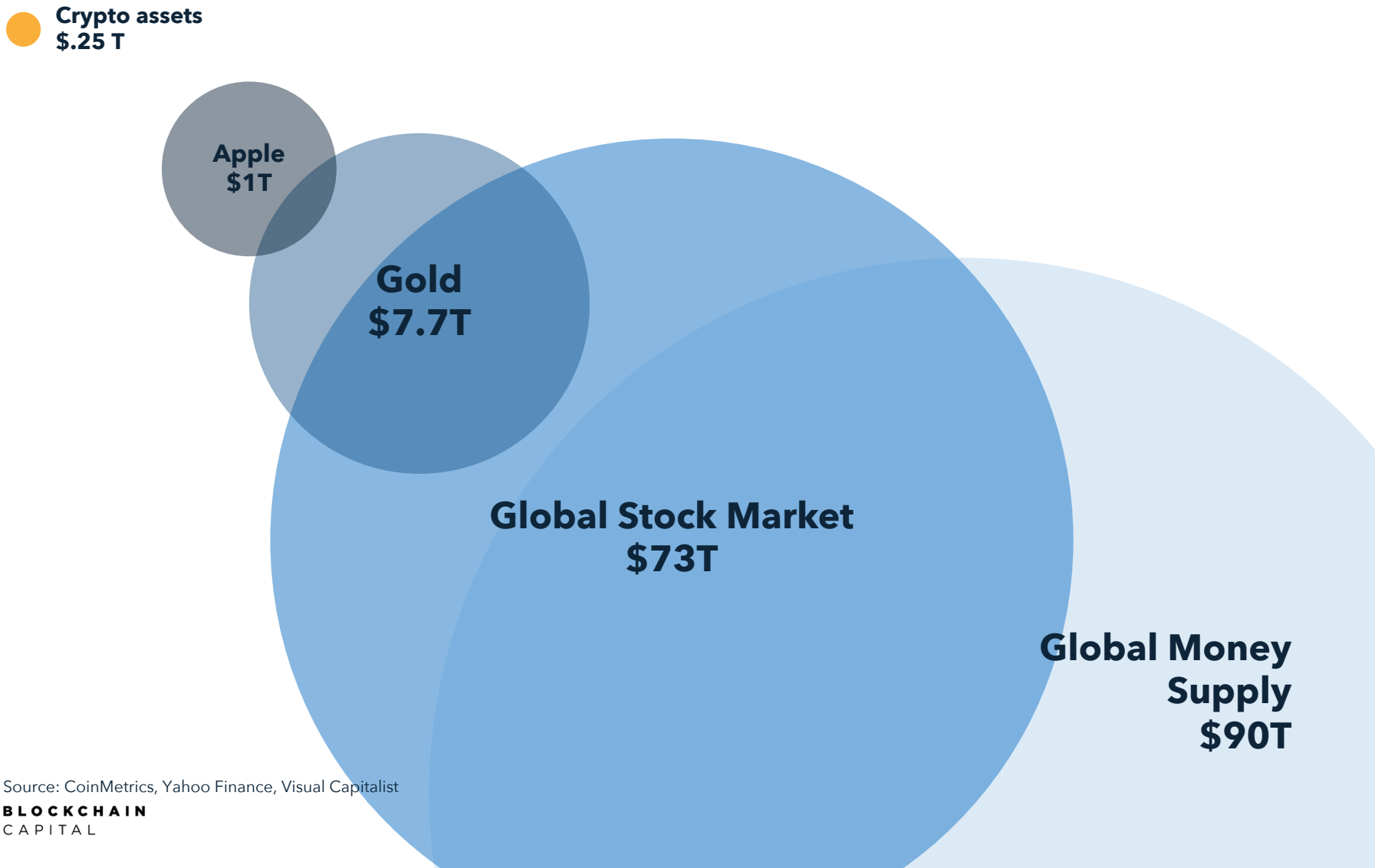
Zooming out on crypto shows we are in the early stages



Source: CoinMetrics, Yahoo Finance, Visual Capitalist

Crypto is still in the early "onboarding phase"

Onboarding Phase | **Opportunities beyond Onboarding**

1980s — *Internet Companies* — *2010s*

'onboarding' companies are typically exchanges and others providing onramps to crypto - as such, it's no surprise that in the current onboarding phase exchanges have been most profitable companies to-date

2010s — *Crypto Companies* — *2030s*

As the number of people 'onboarded' into crypto grows, we can expect more mainstream applications of programmable money to emerge

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Today, protocol development is largely focused on horizontal competition

Present State: Horizontal Competition

$1.8B MKT CAP LAUNCHED
MAJORITY REMAIN UNLAUNCHED

$16B MKT CAP

$133B MKT CAP

Source: CoinMetrics

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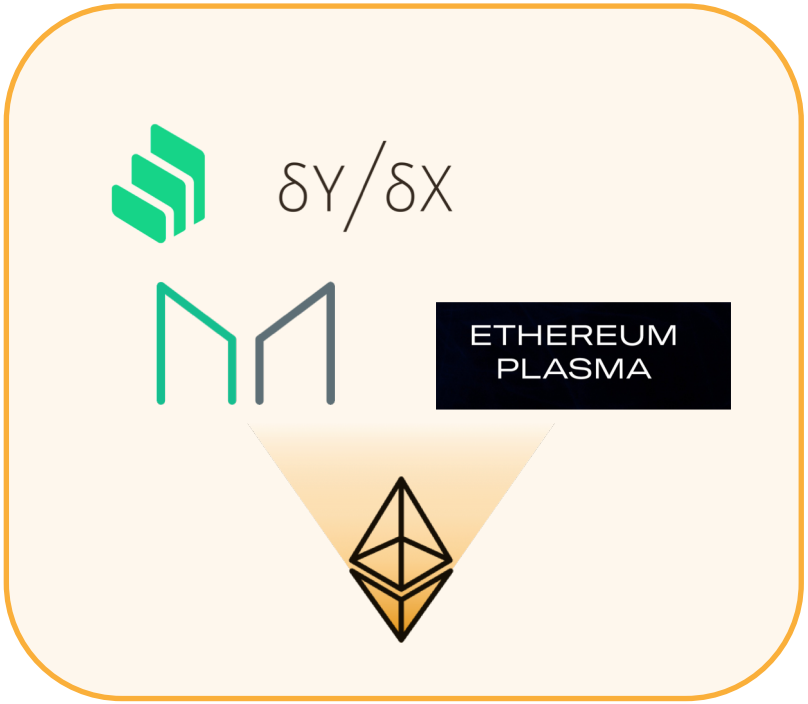
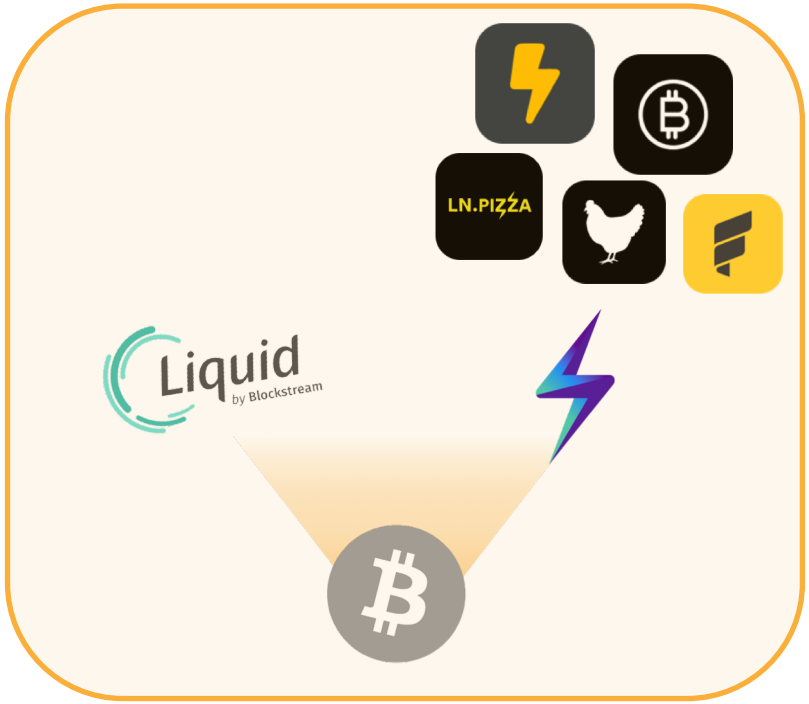
Developer momentum may shift from horizontal competition to vertical construction

Future State: Vertical Construction





As layer 1 further ossifies, development may shift towards vertical construction for a few base layer chains

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As industry reaches inflection point, M&A activity is expected to increase

M&A activity has primarily been limited to internal players…

POLONIEX

seedinvest by CIRCLE

coinbase

xapo Earn

STELLAR

Chain

…but incumbents are quickly devising their blockchain strategies

VISA

Fidelity INVESTMENTS

PayPal

mastercard

Microsoft

aws

M&A activity expected to increase as incumbents lean in and companies demonstrate ability to generate stable cash flows and growth

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Blockchain Capital's bold 2020 predictions

1. A crypto company is acquired for more than $500m
2. Value locked in DeFi hits $5B
3. In the face of competition from China, Libra will receive the green light for a dollar-backed stablecoin
4. A federal judge rules against the SEC in a crypto case
5. Not a single 2020 L1 network launch achieves "top 10" status, as defined by network value
6. USDC sees 300%+ growth (as measured by transaction value, issuance, market cap and trading volume)
7. Demand for Bitcoin transactions drives fees to exceed $100, catalyzing scaling up the stack
8. FinCEN / FATF hold stablecoins to a stricter standard than paper cash by requiring broad application of the travel rule
9. McAfee loses his bet and eats his…
10. KYC / AML becomes the primary regulatory battleground for DeFi
11. Privacy coins are de-listed from major exchanges
12. Bitcoin price blows past all-time high

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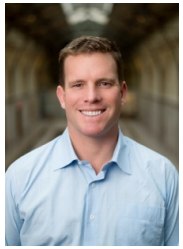
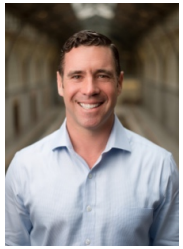
About Blockchain Capital

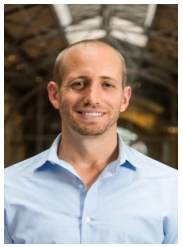
INVESTMENT TEAM



Brad Stephens
Co-Founder and Managing Partner



Bart Stephens
Co-Founder & Managing Partner



Spencer Bogart
General Partner

Ben Davenport
Venture Partner

Jimmy Song
Bitcoin Fellow

H. Joshua Rivera
General Counsel & Chief Compliance Officer

Kinjal Shah
Senior Associate

Aleks Larsen
Associate

Derek Hsue
Analyst

AT A GLANCE

- 80+ portfolio companies
- $300M+ in assets under management
- 9 dedicated investment professionals
- 4 funds to date

PORTFOLIO SNAPSHOT

ripple
coinbase
kraken
BitGo
BISON TRAILS
ANCHORAGE
RADAR
SECURITIZE
TRM
Bitwise

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@blockchaincap
contact@blockchaincapital.com

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